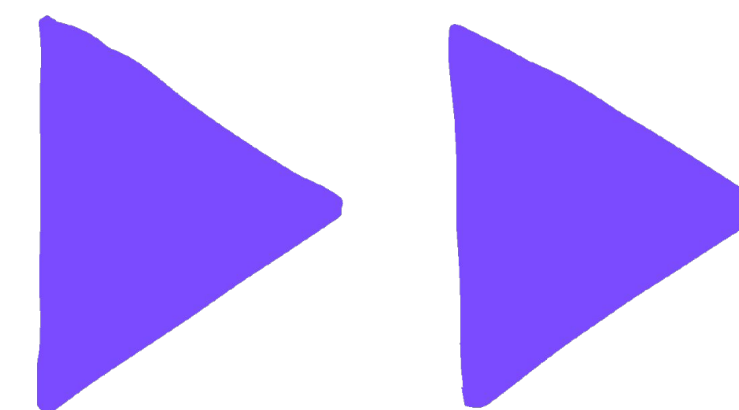
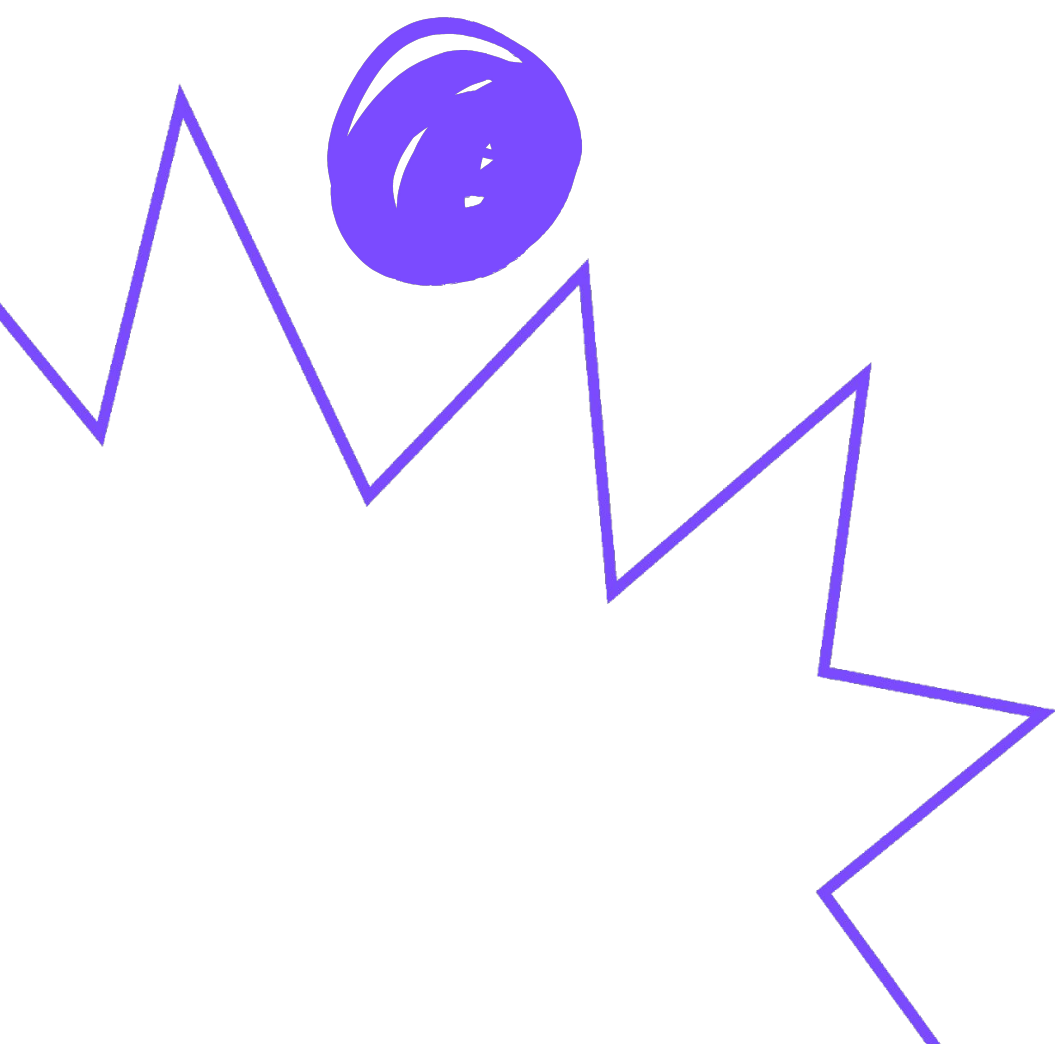




AI engineering tools 3



Это полностью прикладной курс.





Шаповалов Александр

aTCL @ Internal automation

- Вижу, как медленно внедряется AI в бигтехе
- tg: @rostish

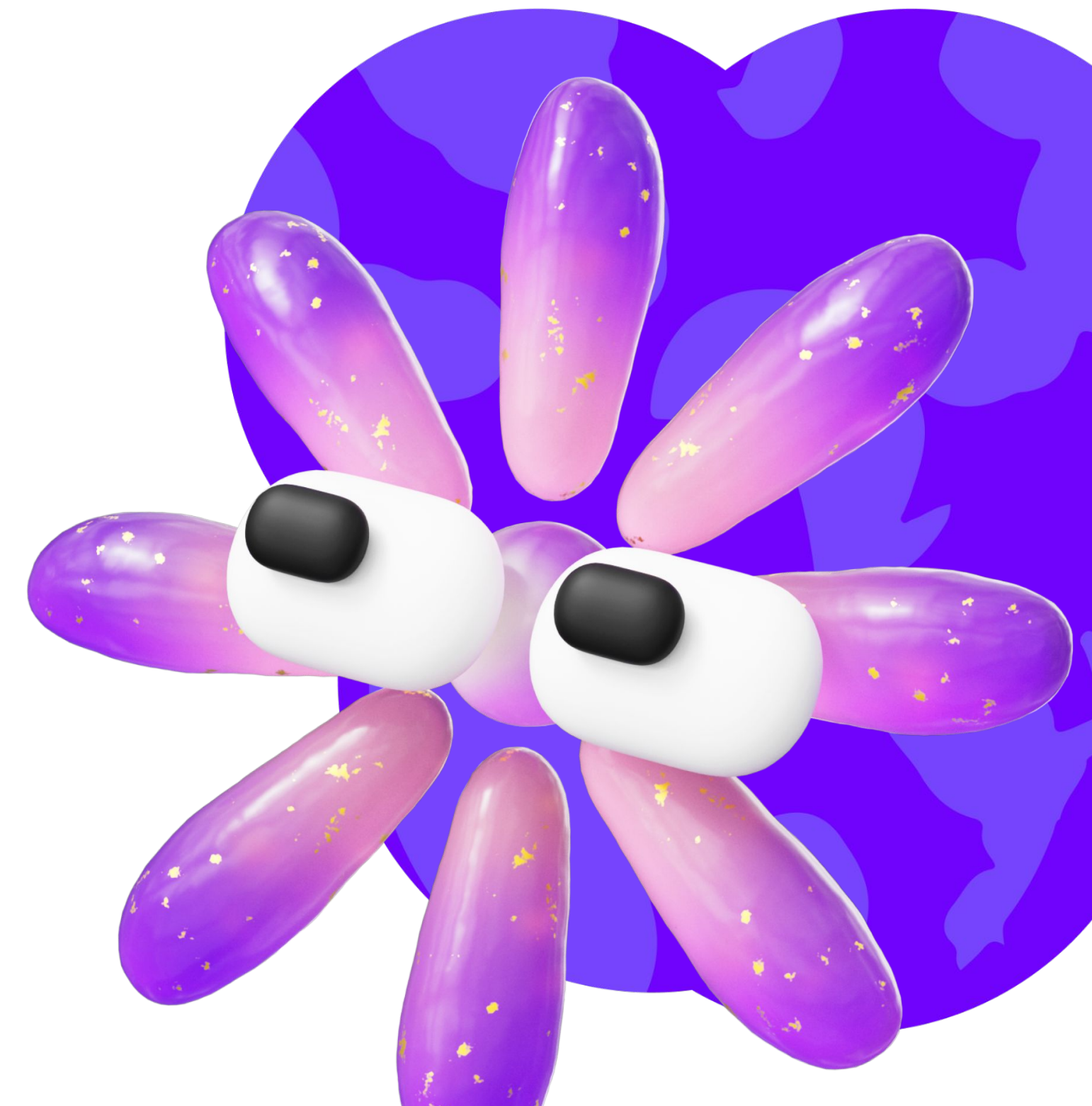
Условия успешного прохождения дисциплины

1. Посещение лекционных и лабораторных занятий — min 50% от всего количества.
2. Активное участие на занятиях — доп. баллы
3. Соблюдение дедлайнов сдачи лабораторных работ — можно сдать позже дедлайна на 3 дня и получить штраф -1 балл. Пересдать лабу можно в течение недели после проверки.
4. Самостоятельные работы выполняются самостоятельно



03

Разработка с использованием coding agents



Сегодня узнаете

1. Эволюция агентских систем
2. Тренды на 2026 - 2030
3. Coding agents



R1. Эволюция агентских систем

1. Понятие агента

2. LLM powered agentic system

2



2

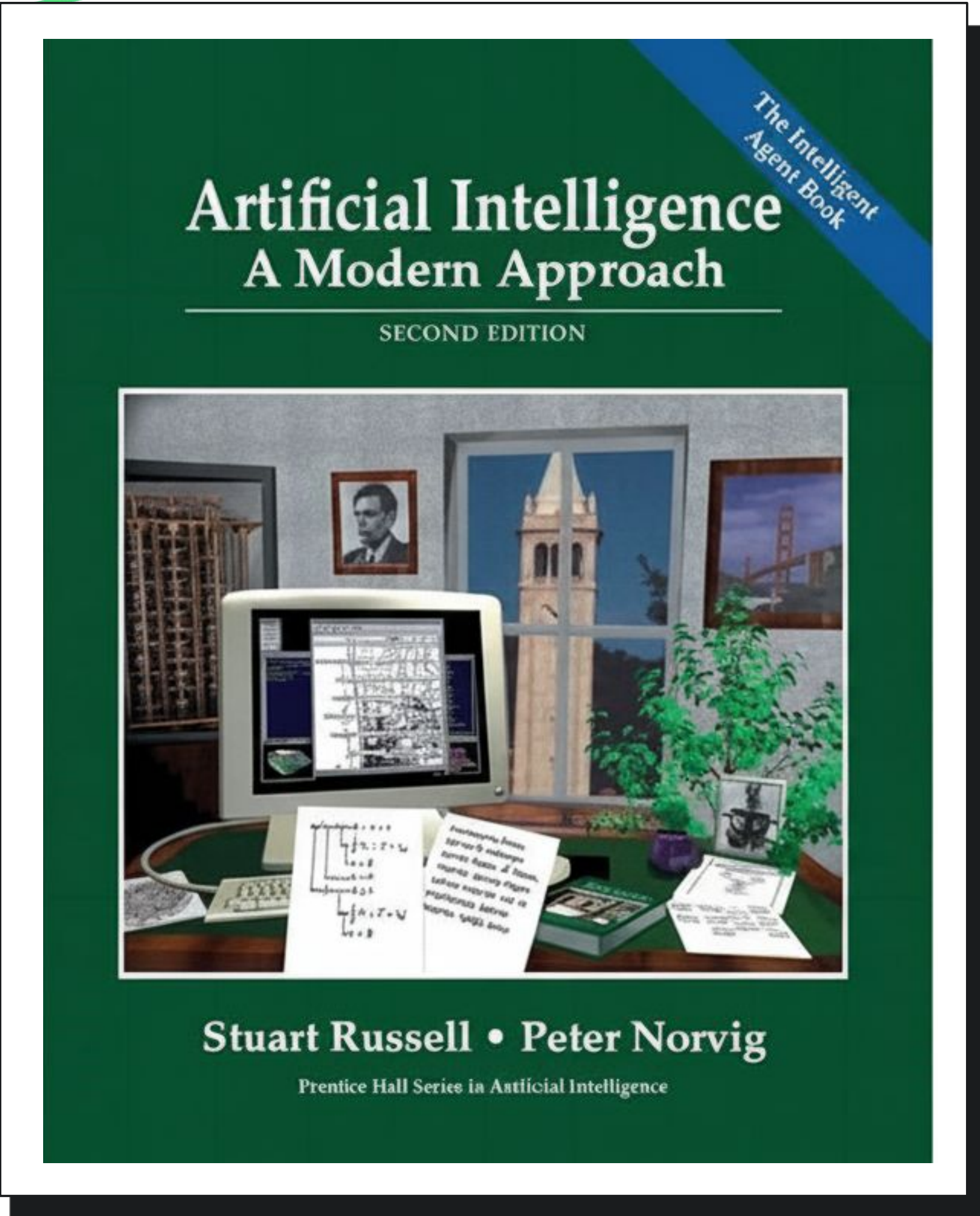
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4

Стюарт Рассел и Питер Норвиг



Стюарт Рассел и Питер Норвиг



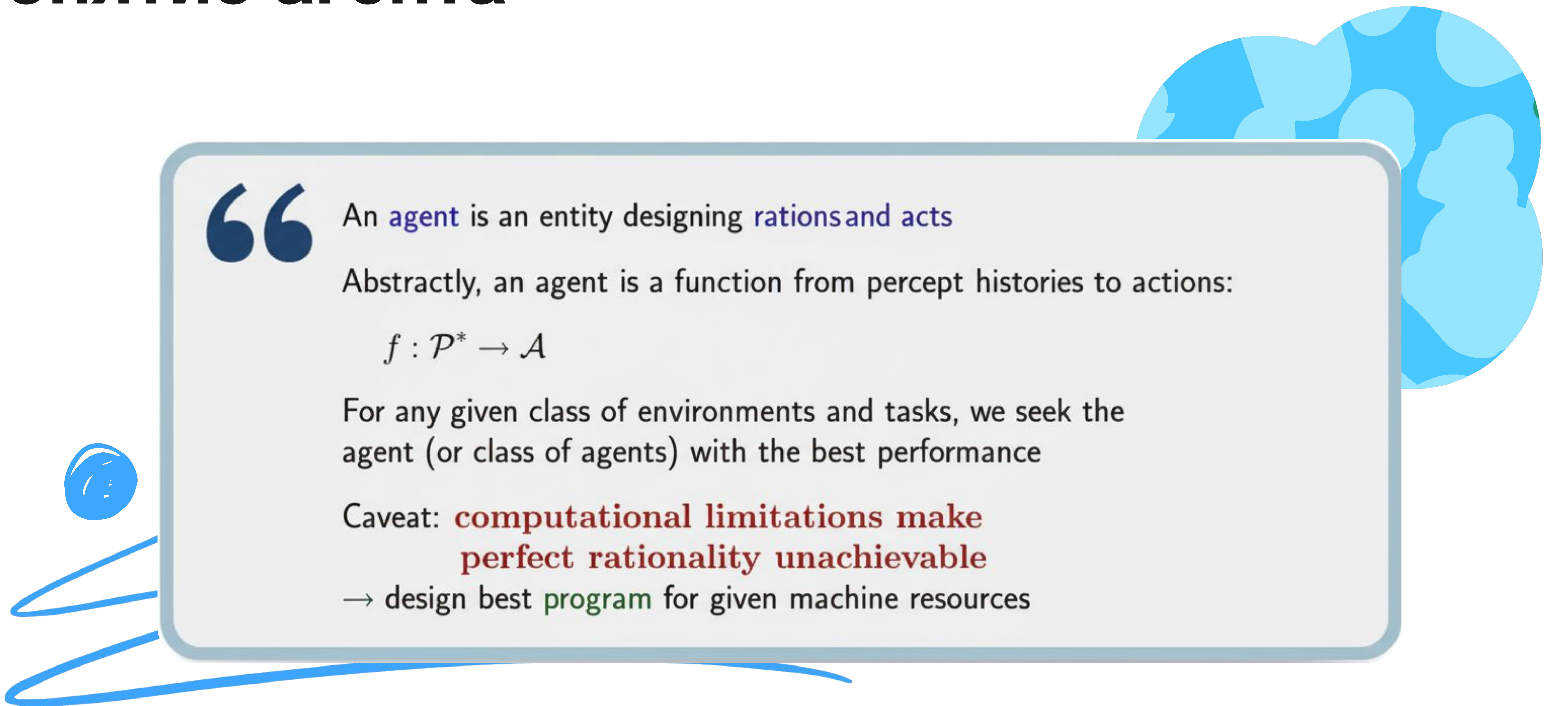
Стюарт Рассел и Питер Норвиг



<https://inst.eecs.berkeley.edu/~cs188/sp26/assets/lectures/>



Понятие агента



“

An **agent** is an entity designing **rations** and **acts**

Abstractly, an agent is a function from percept histories to actions:

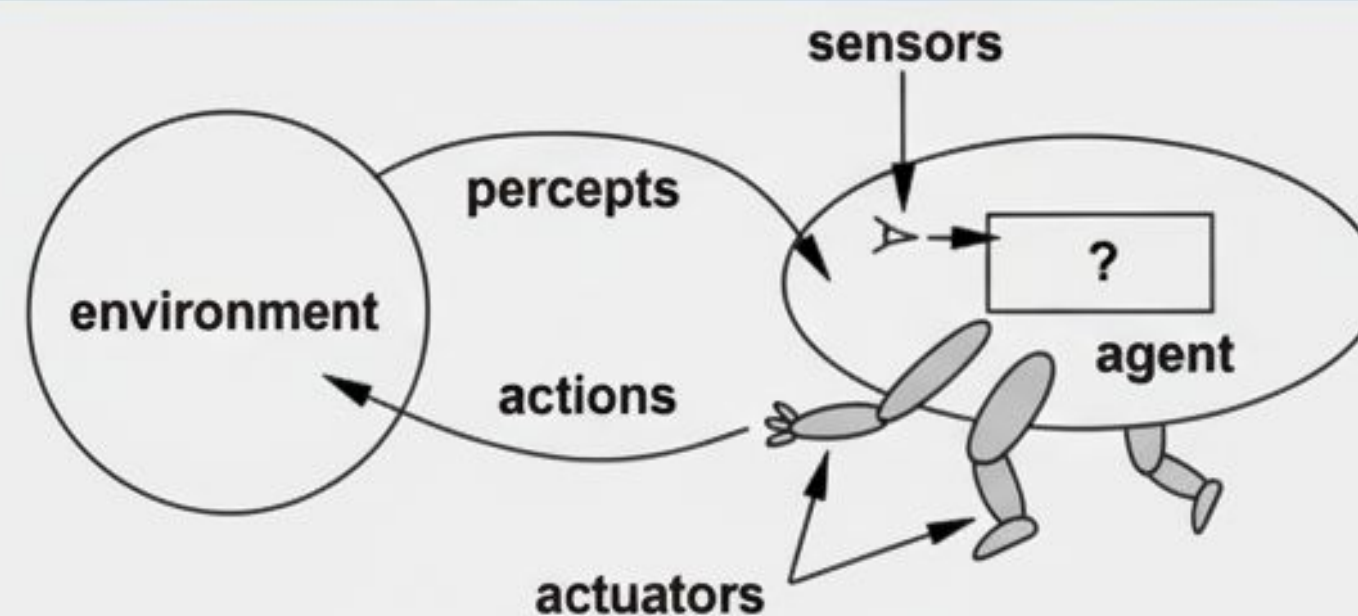
$$f : \mathcal{P}^* \rightarrow \mathcal{A}$$

For any given class of environments and tasks, we seek the agent (or class of agents) with the best performance

Caveat: **computational limitations make perfect rationality unachievable**
→ design best **program** for given machine resources

Стюарт Рассел и Питер Норвиг

“



Abstractly, an agent as function, robots, thermostats, etc.

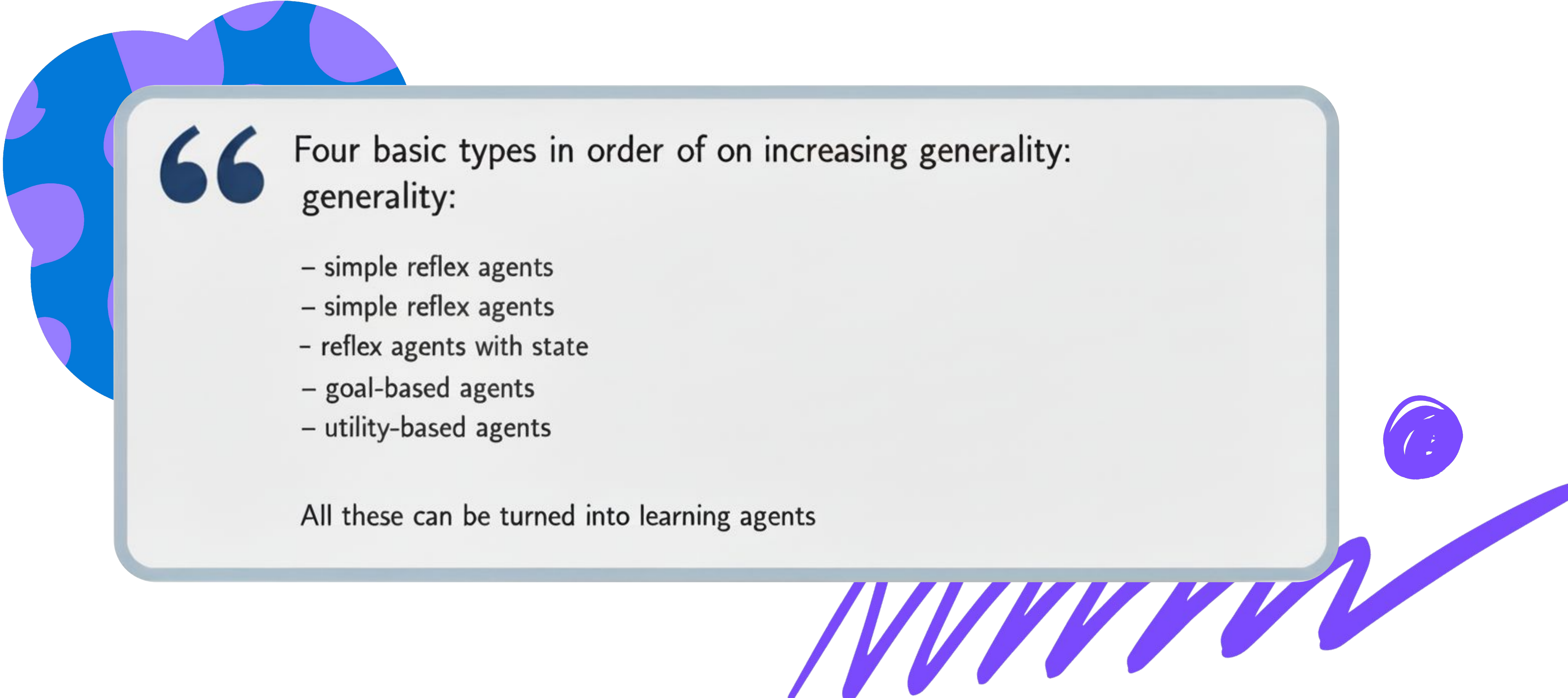
The **agent function** maps percept histories to actions:

$$f : \mathcal{P}^* \rightarrow \mathcal{A}$$

The **agent program** runs on the physical **architecture** to produce f



Стюарт Рассел и Питер Норвиг



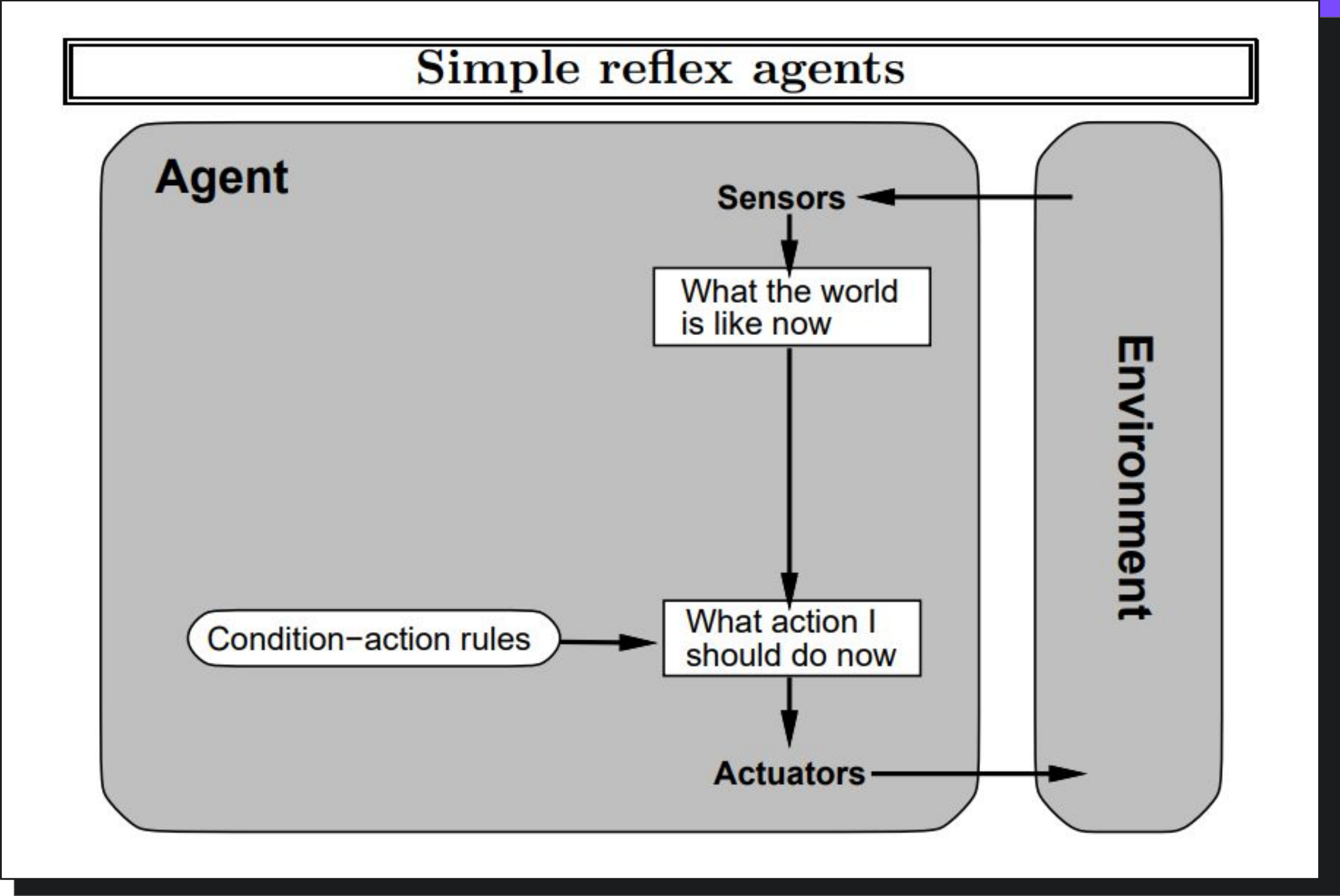
“

Four basic types in order of on increasing generality:
generality:

- simple reflex agents
- simple reflex agents
- reflex agents with state
- goal-based agents
- utility-based agents

All these can be turned into learning agents

Типы агентов

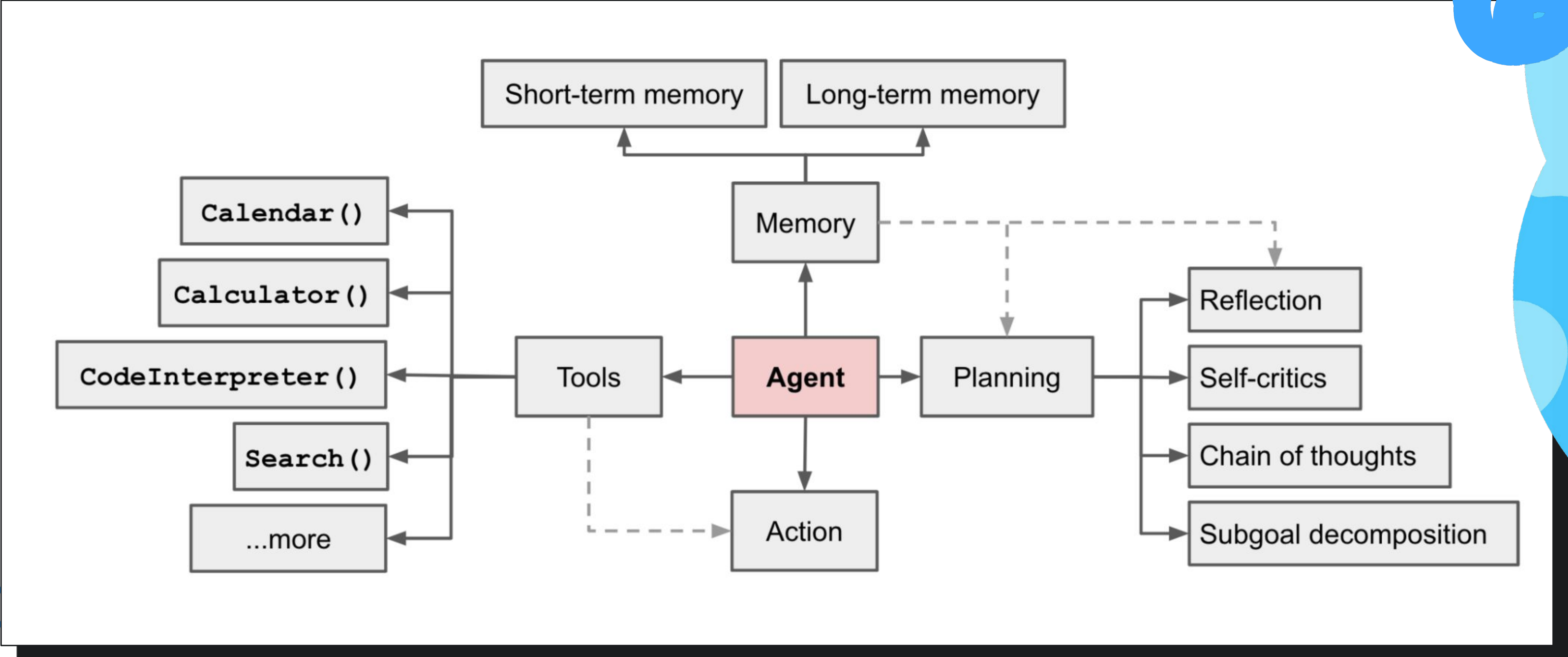




Придумайте по одному агенту.



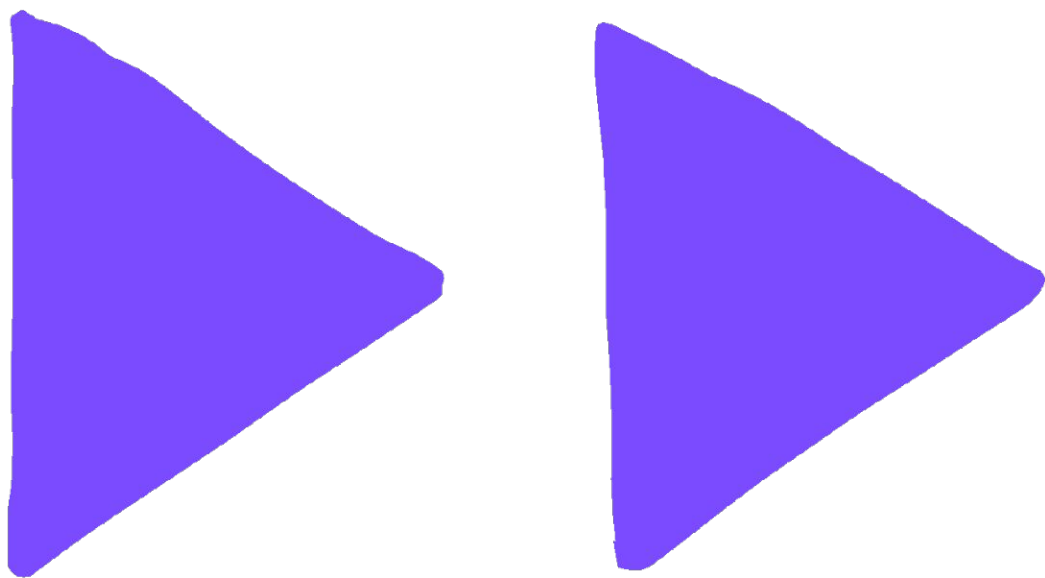
LLM powered agentic system



Planning

Для планирования используются знакомые вам техники:

1. CoT
2. ToT
3. PDDL



Planning

PDDL (planning domain definition language)

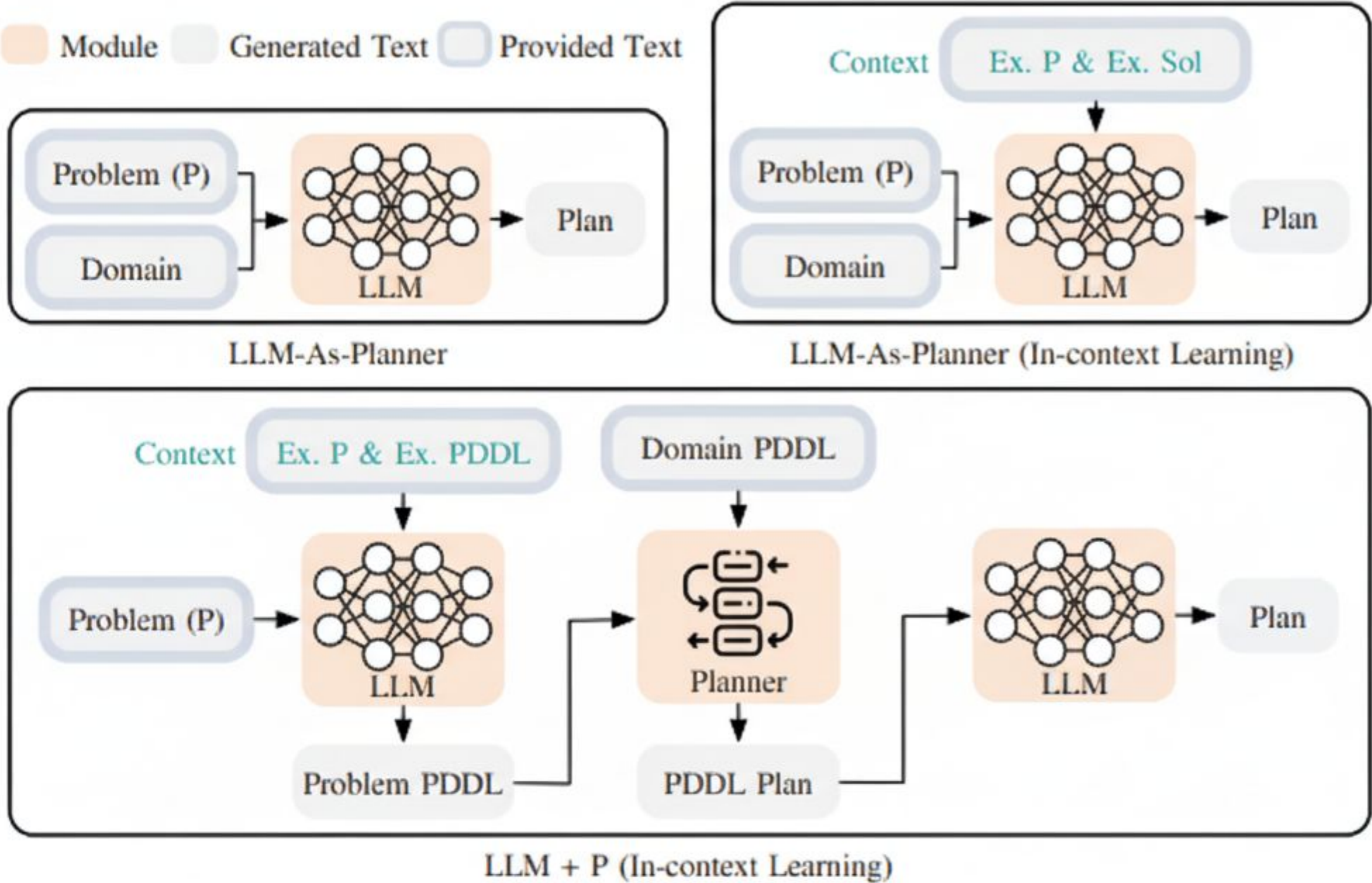
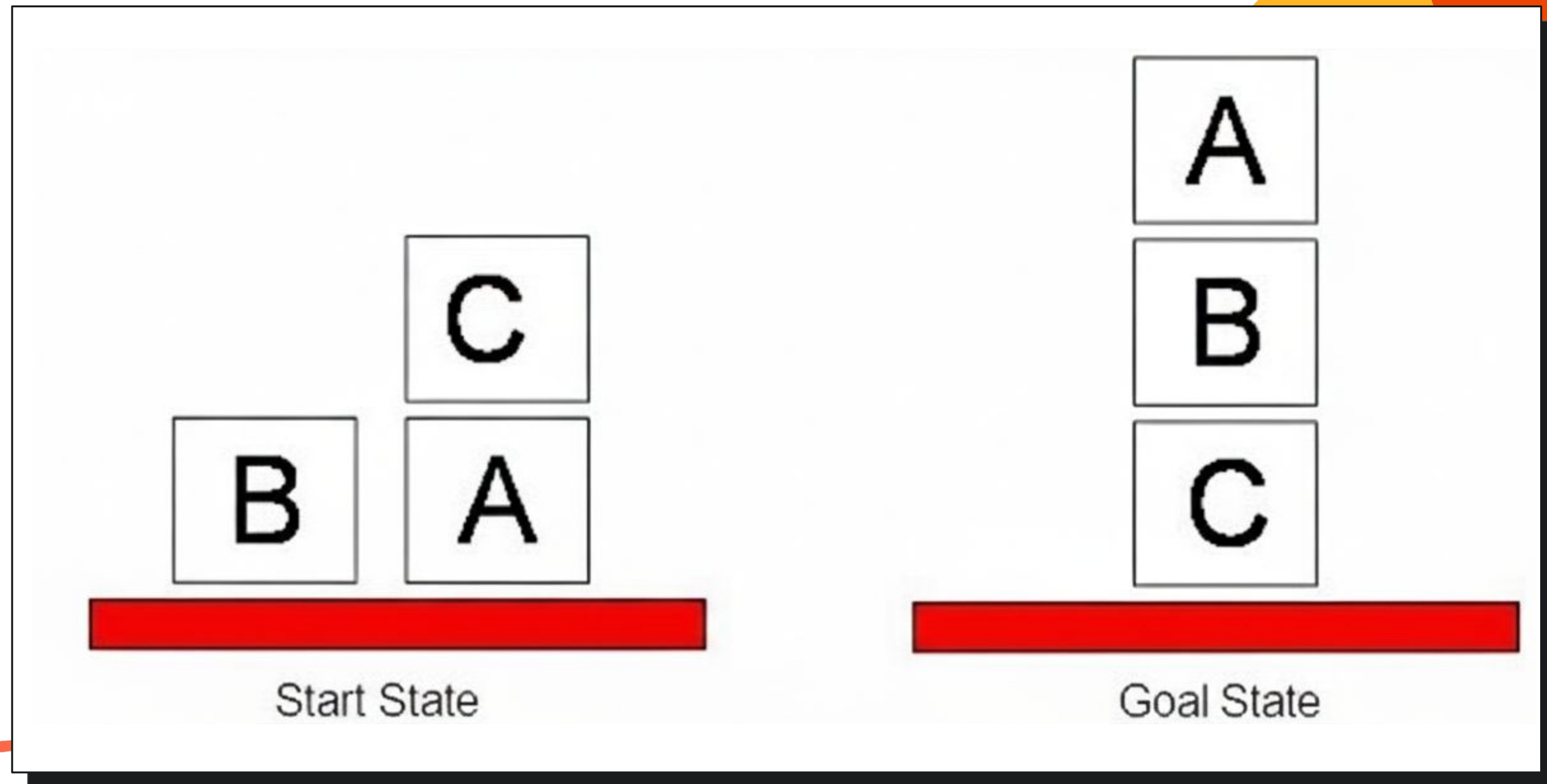


Fig. 1: LLM+P makes use of a large language model (LLM) to produce the PDDL description of the given problem, then leverages a classical planner for finding an *optimal* plan, then translates the raw plan back to natural language using the LLM again.

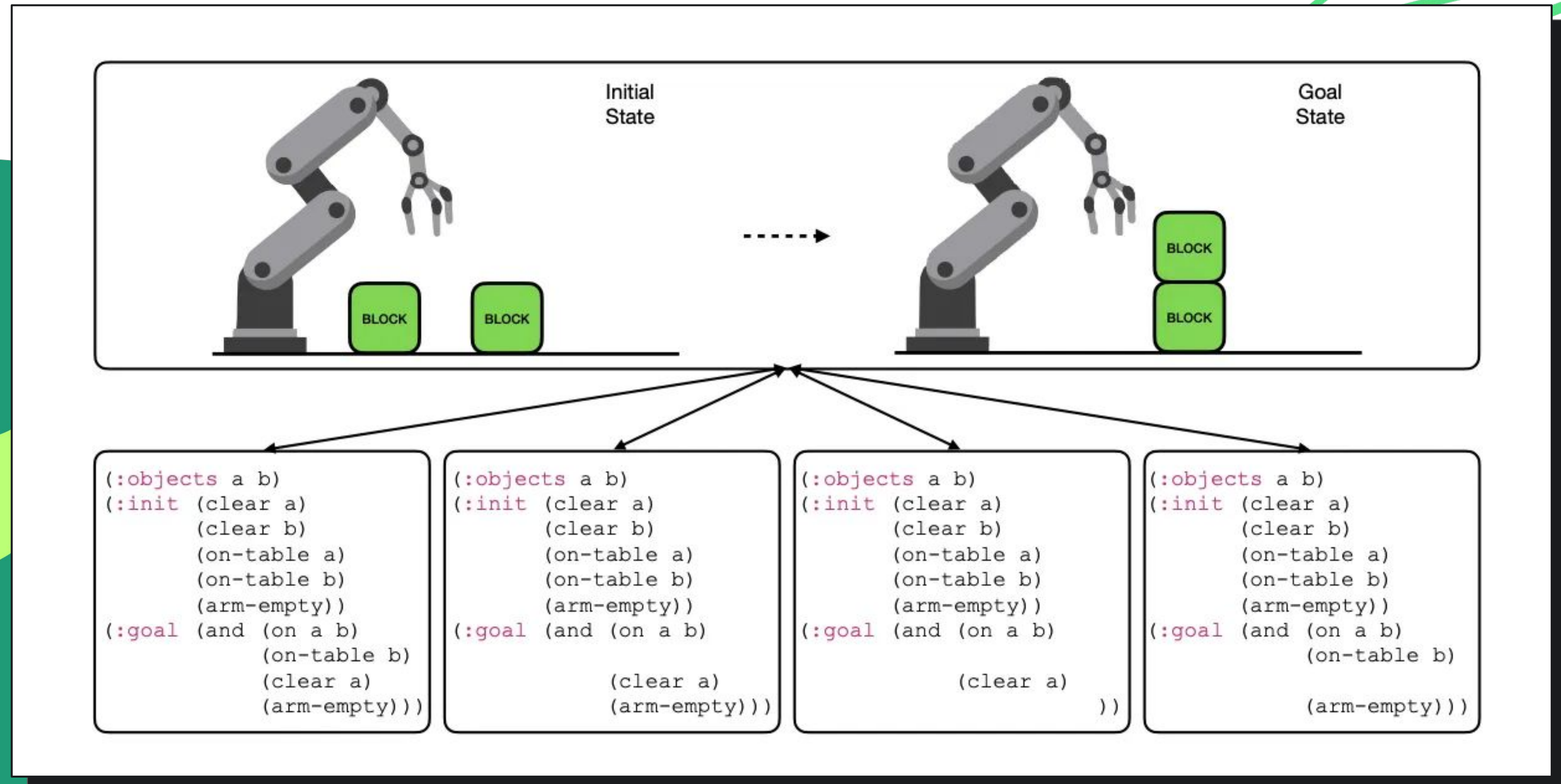
Planning

PDDL (planning domain definition language)



Planning

PDDL (planning domain definition language)



Planning

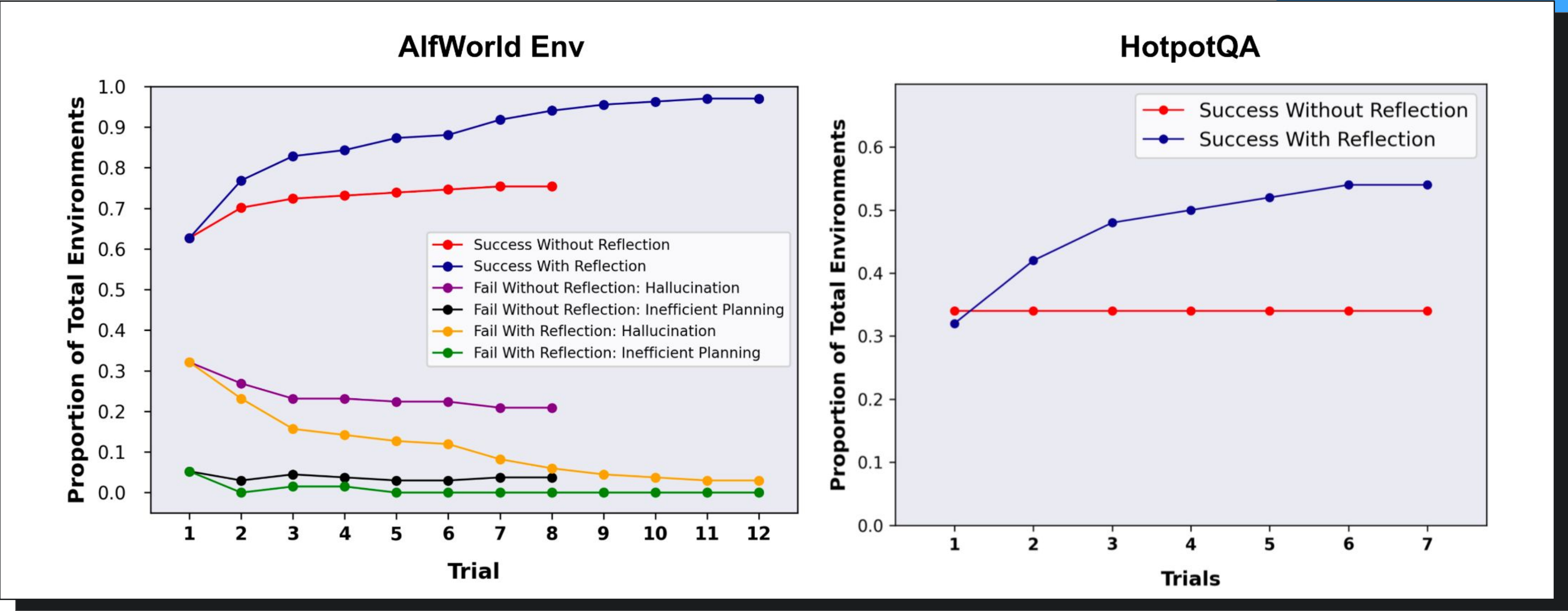
Self-reflection

В качестве инструмента используется паттерн ReAct.

Структура промта:

“Thought: ...
Action: ...
Observation: ...
... (Repeated many times)”

Planning

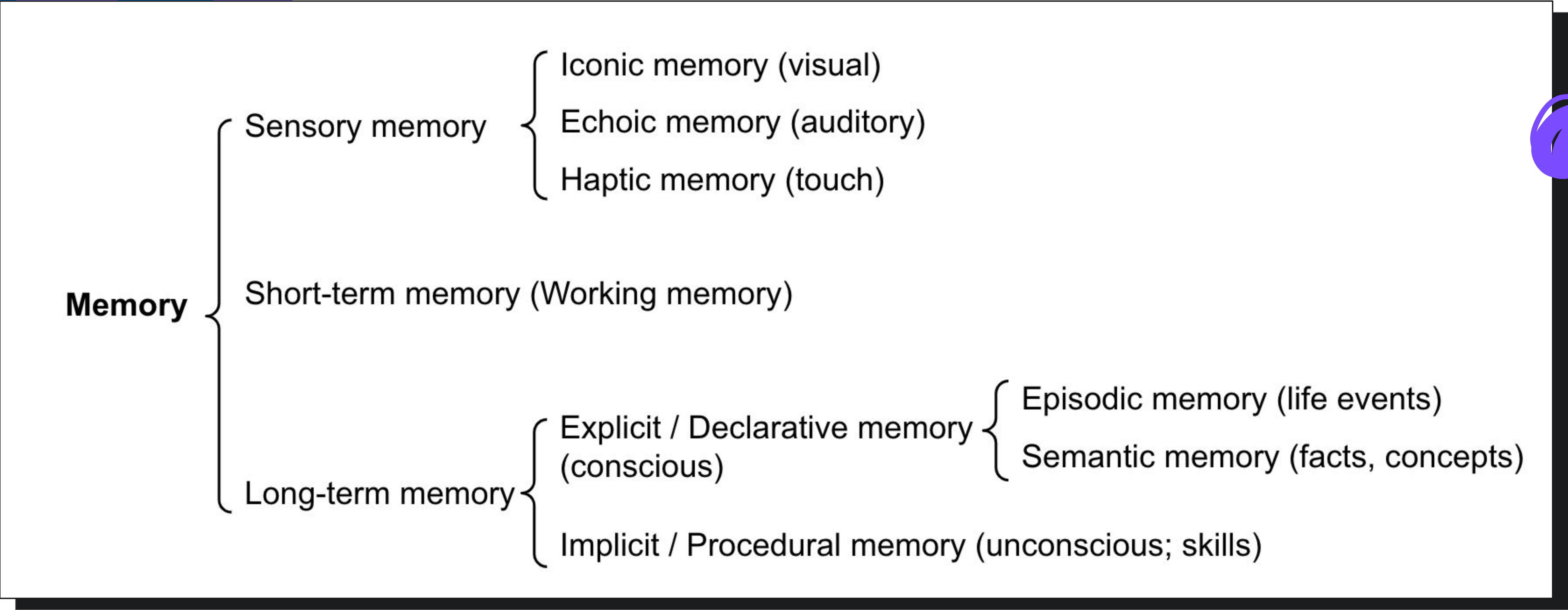


Planning

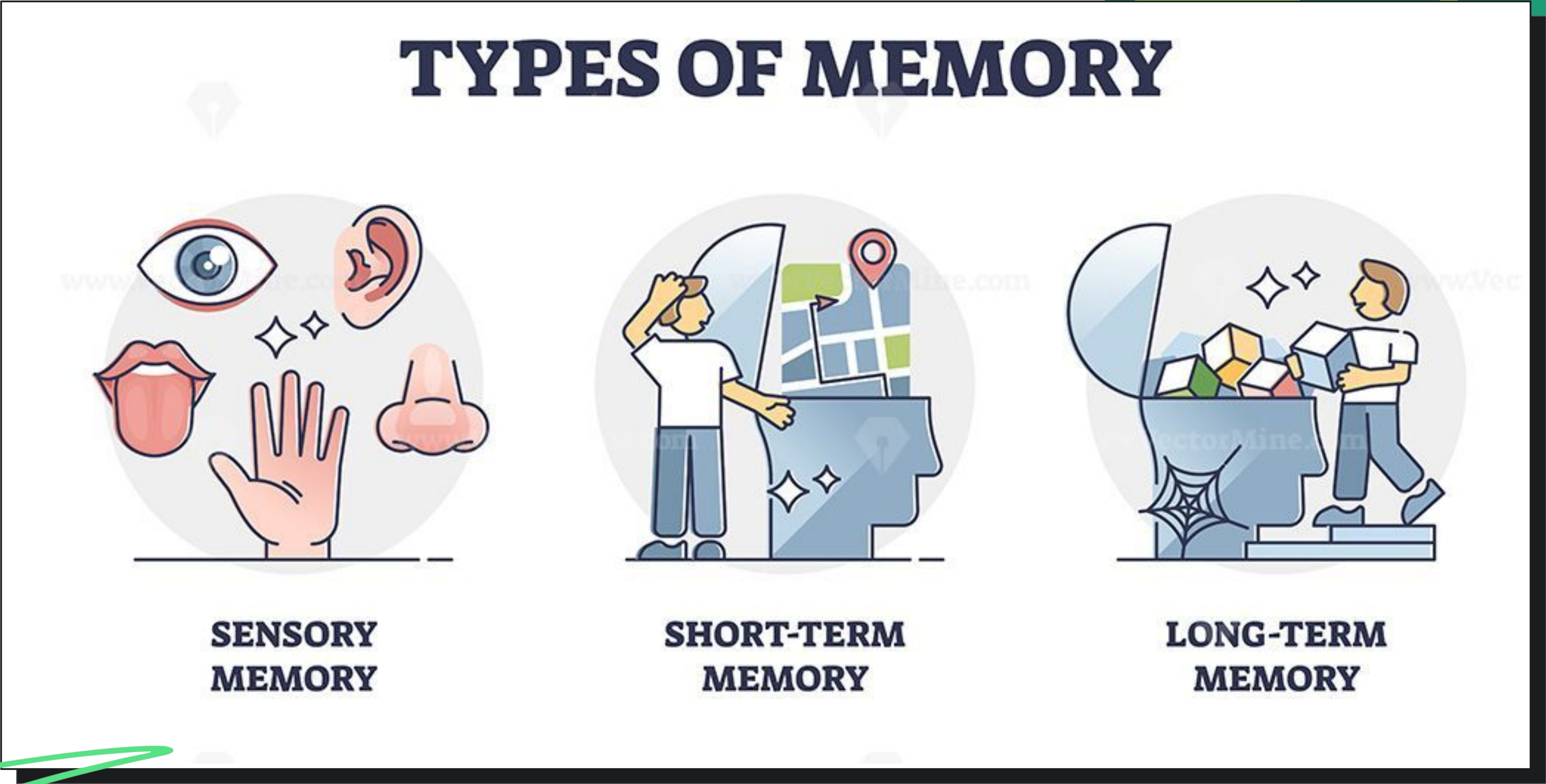
**Ничего не
напоминает?**



Memory

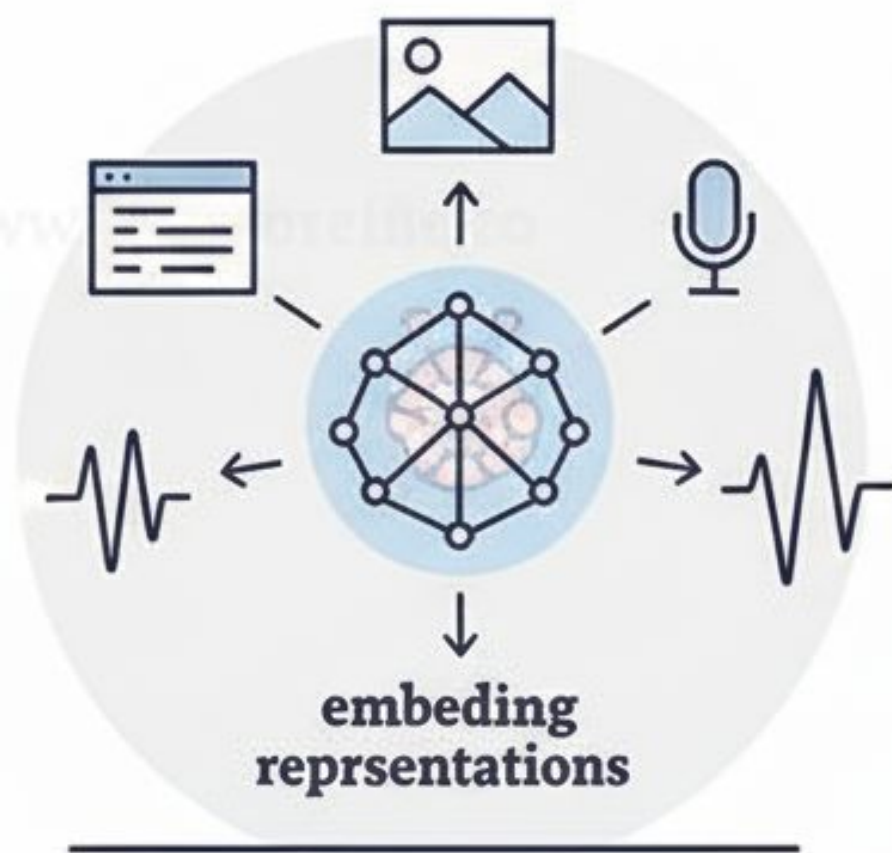


Memory

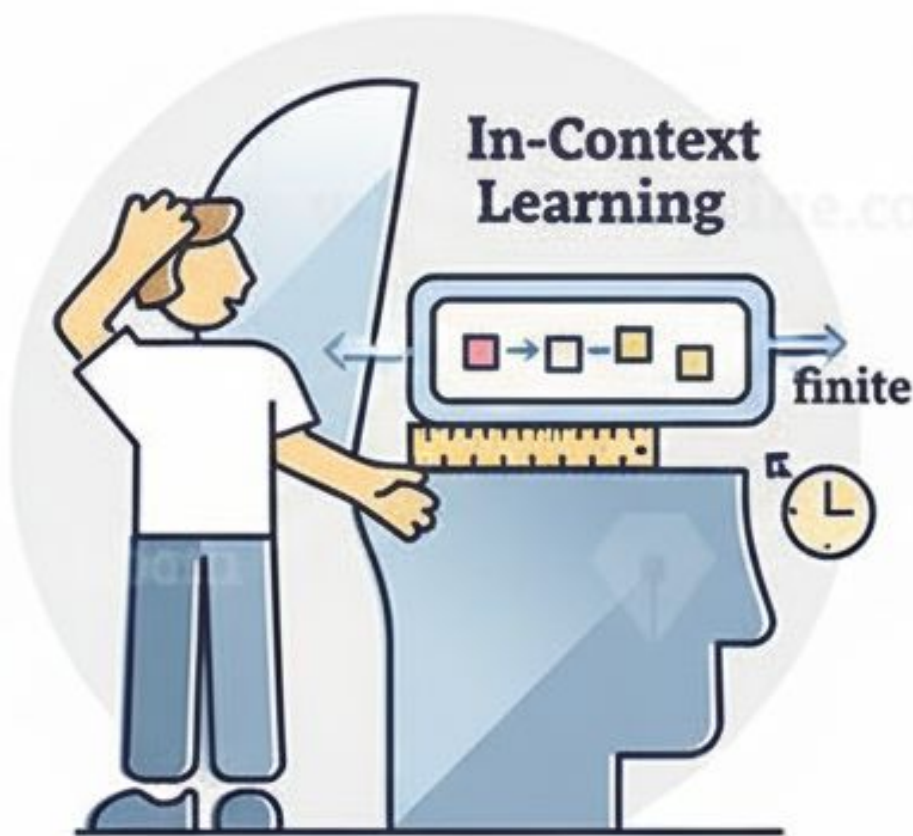


Memory

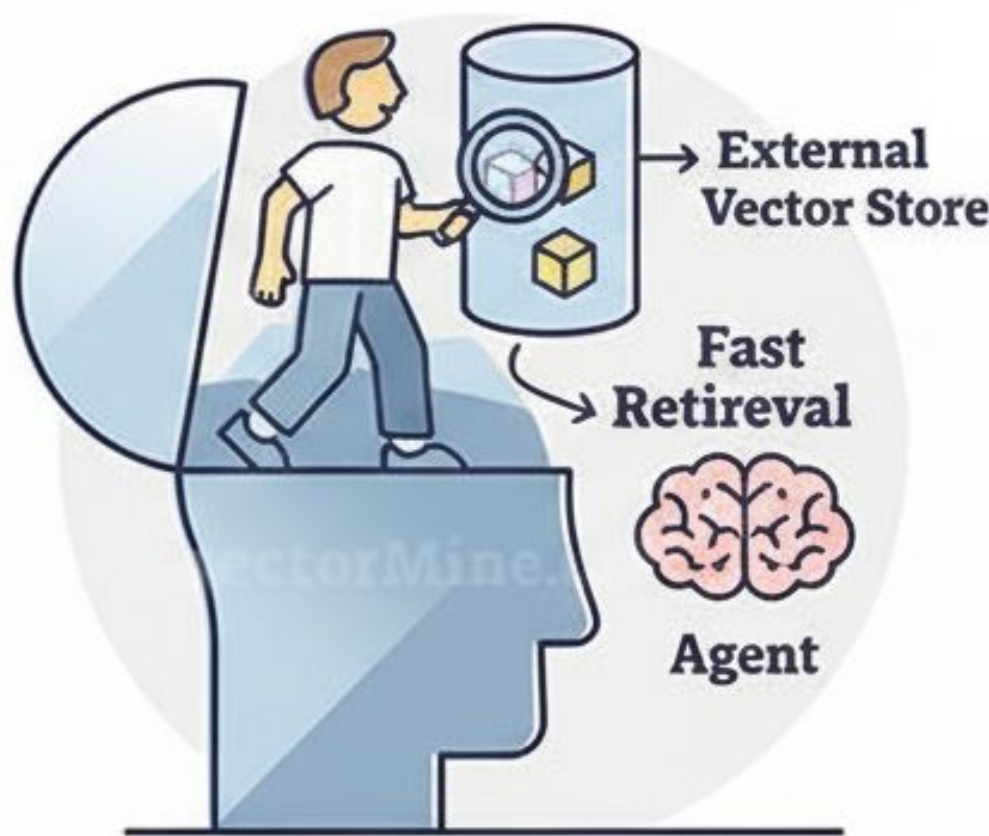
TYPES OF AI MEMORY



SENSORY
MEMORY

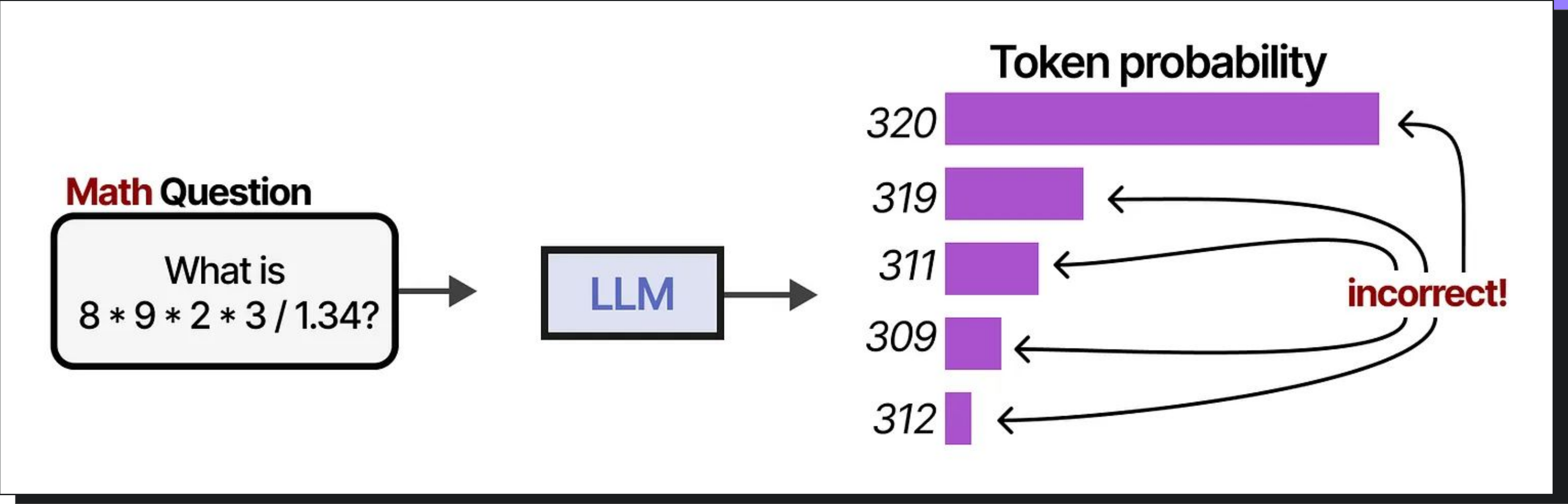


SHORT-TERM
MEMORY

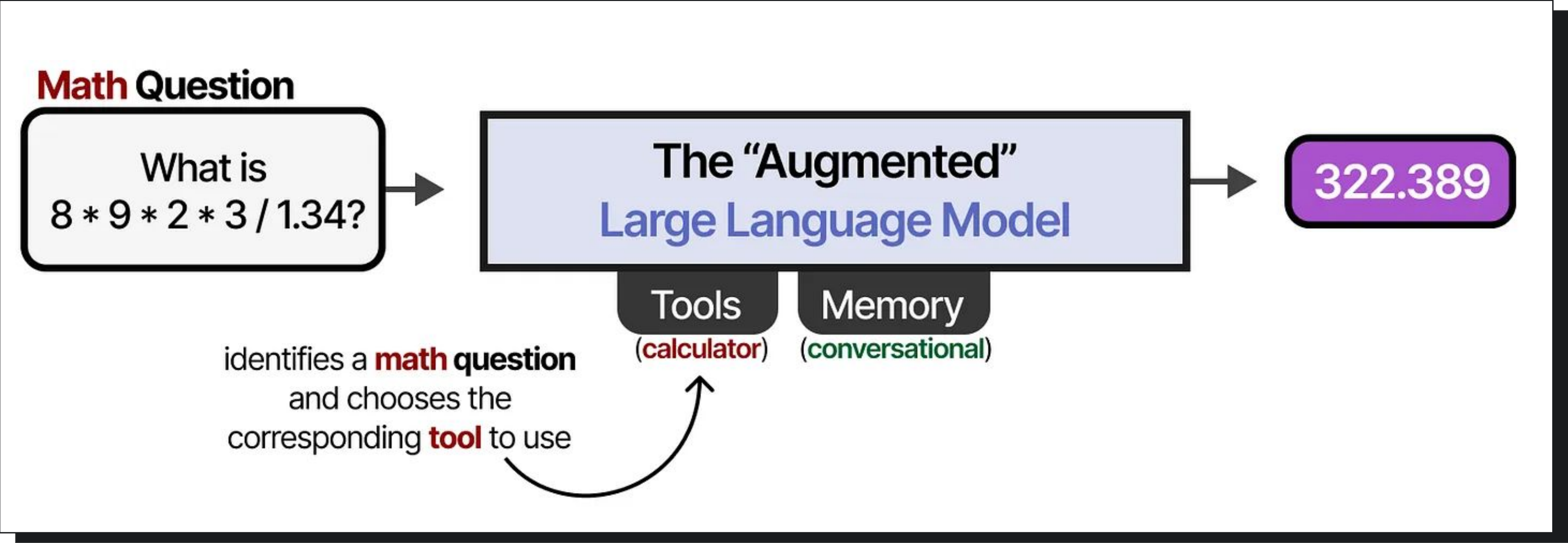


LONG-TERM
MEMORY

Tools use



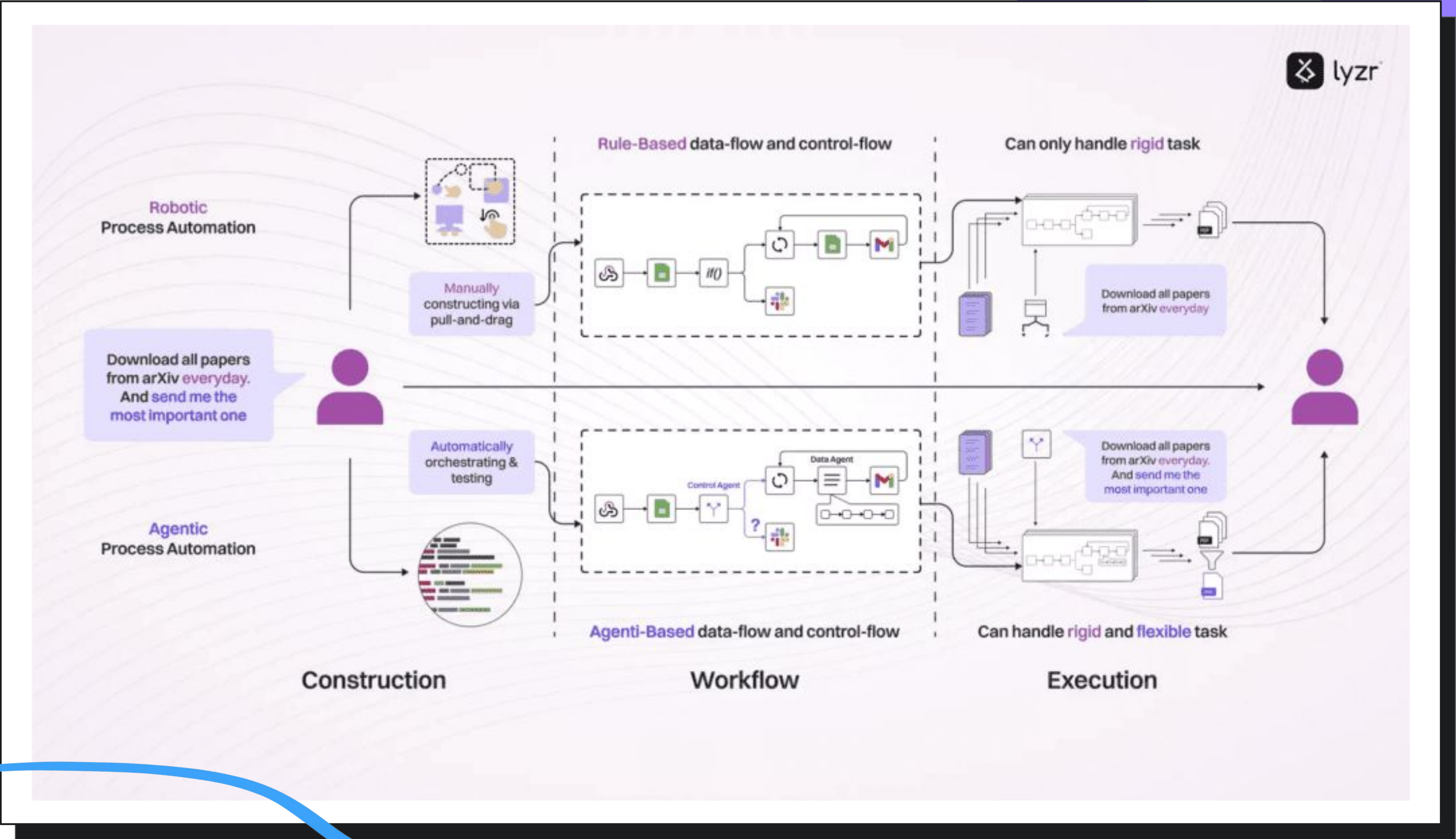
Tools use



Напишем микро агента с калькулятором?



Robotic vs Agentic подходы



Про agent системы



Maarten Grootendorst
Author "Hands-On Large
Language Models"

<https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-llm-agents>



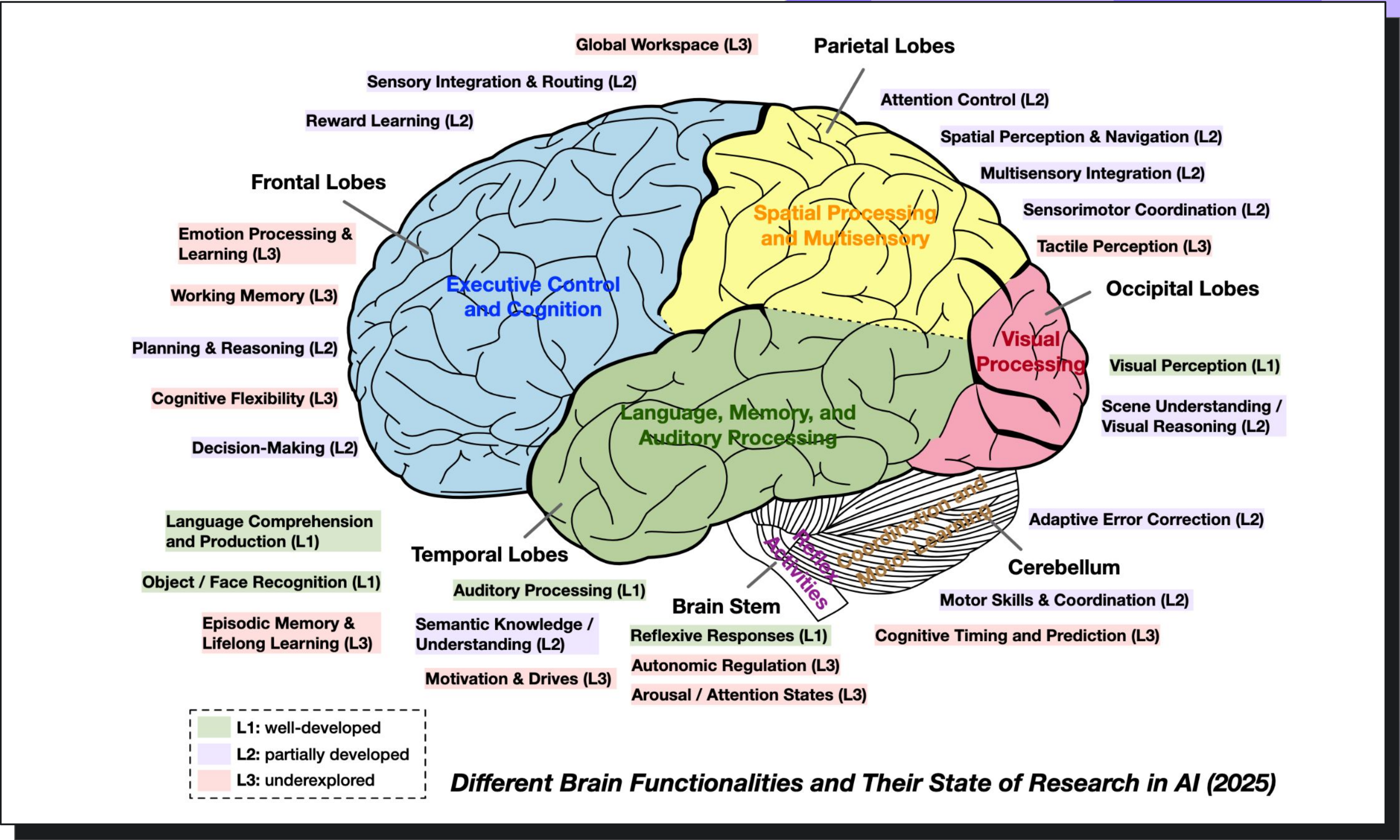
Lilian Weng (OpenAI)
Глава Safety Systems в
OpenAI

<https://lilianweng.github.io/posts/2023-06-23-agent/>

Arxiv

<https://arxiv.org/pdf/2304.11477>





R2. Тренды на 2026 - 2030

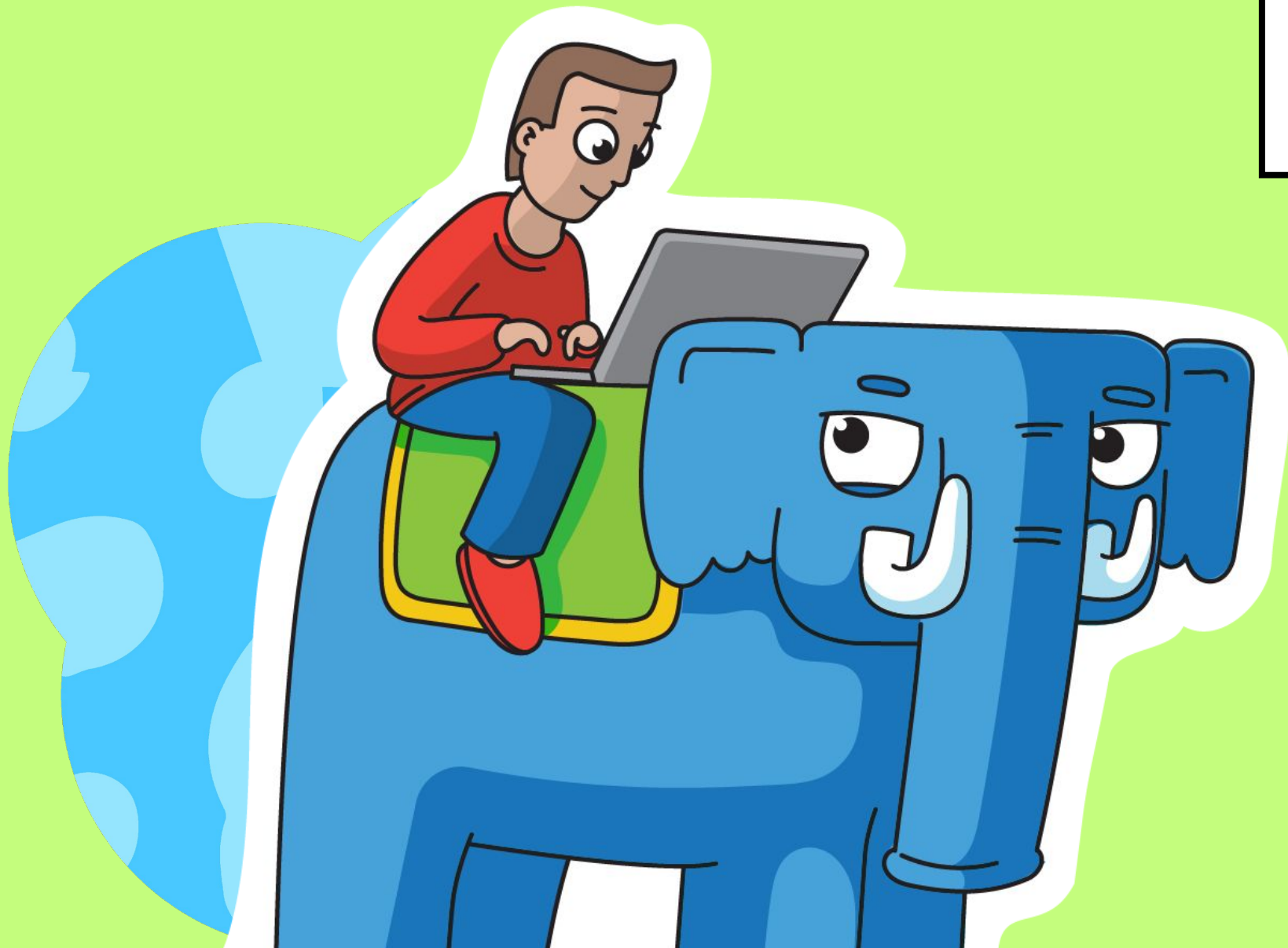
4

1. Исследование Gartner

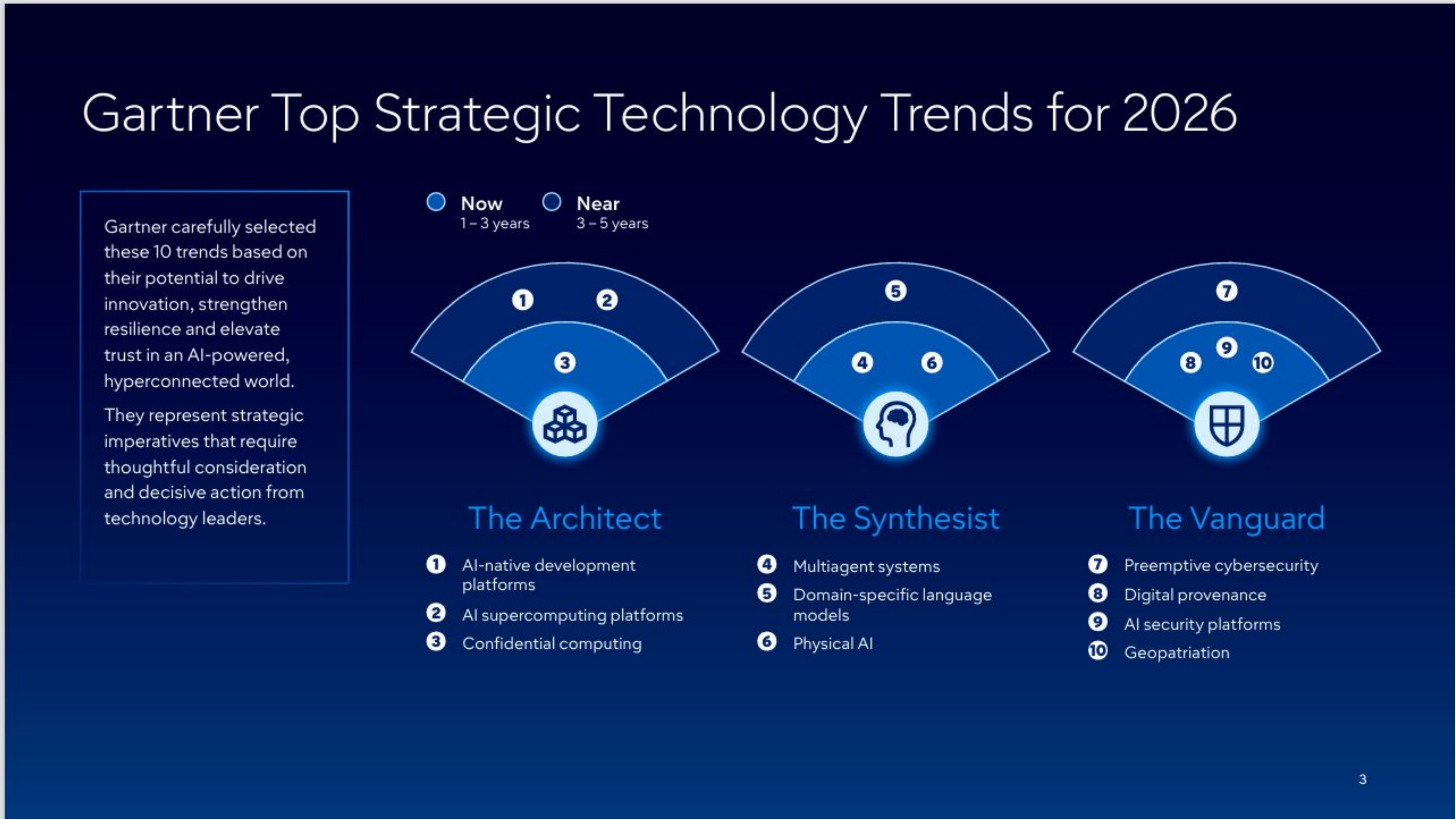
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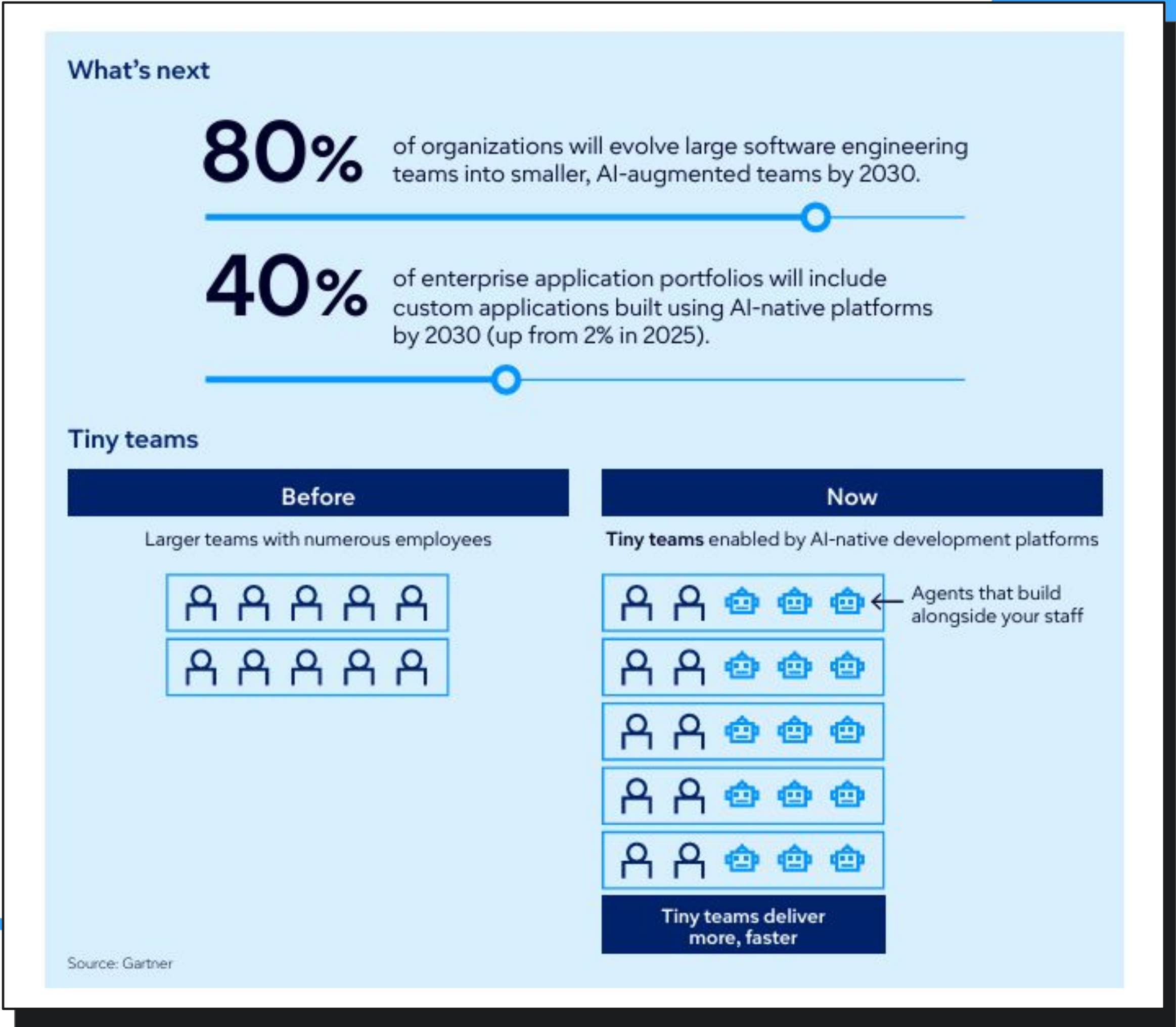
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Технологические тренды к 2026 году



Структура команд к 2030 году



Как будет достигаться эффект

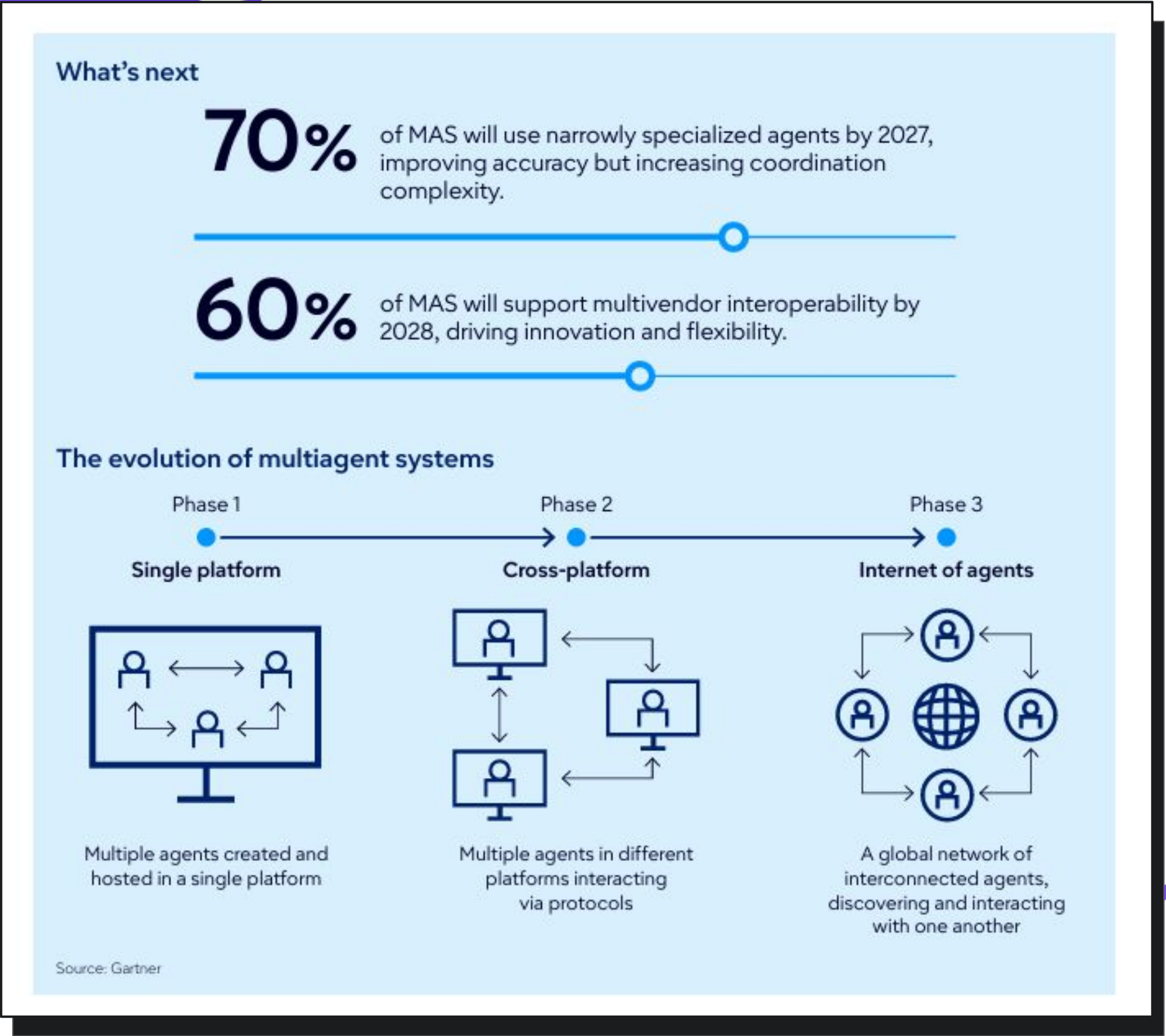
Action plan to boost speed, save costs and spark innovation

Steps	1 Establish a platform team	2 Implement security guardrails	3 Pilot AI-native development	4 Adopt an AI-first mindset	5 Upskill and enable teams
Expected outcome	Centralized oversight ensures consistent standards and governance.	Reduced risk of insecure or noncompliant code.	Quick wins that demonstrate value and build confidence.	Accelerated delivery and improved innovation capacity.	Broader adoption and effective collaboration.
Action	Form a dedicated team to manage AI-native platforms and select AI models.	Integrate AI governance platforms for code review and compliance checks.	Start with low-risk projects to validate productivity gains.	Prioritize AI-native tools for new development initiatives.	Train developers and business partners on prompt engineering and governance.

Key players to support implementation success

CIO	IT partners	Business partners
Partner: Define AI-first strategy and governance framework. Collaborate: Align platform capabilities with business priorities. Govern: Ensure compliance and security guardrails for AI-native development.	Platform engineering: Manage AI-native tools, integrations and performance. Security: Implement AI governance for code review and risk management. Procurement: Evaluate and select AI-native platform vendors and services.	Product owners: Provide domain expertise and validate AI-driven solutions. Finance: Align funding models to support AI-native development initiatives.

Развитие MAS в 2026- 2030 годах



Как и какой будет достигаться эффект

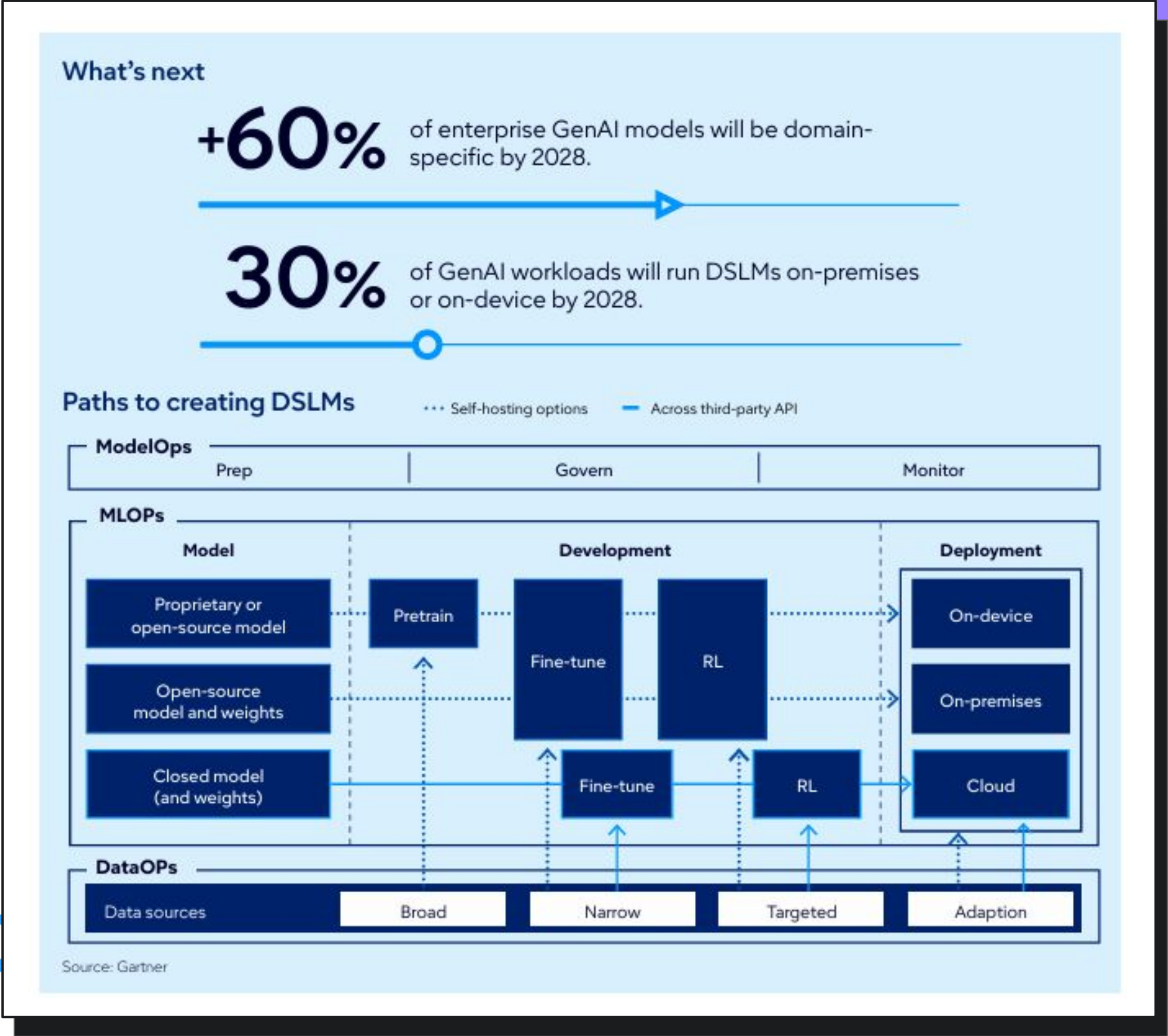
Action plan to drive modular automation and seamless integration

Steps	1 Identify high-value use cases	2 Design modular agents	3 Implement governance and observability	4 Adopt interoperability standards	5 Upskill teams
Expected outcome	Measurable impact and faster adoption.	Improved reliability and scalability.	Reduced risk and better control.	Future-proof MAS investments.	Effective deployment and risk mitigation.
Action	Start with well-defined workflows for MAS pilots.	Build specialized agents instead of monolithic solutions.	Apply strong API governance and monitoring tools.	Use emerging protocols for multivendor agent collaboration.	Train staff on MAS frameworks and change management.

Key players to support implementation success

CIO	IT partners	Business partners
Define MAS strategy for high-value workflows and align with business priorities. Establish governance for agent interoperability, security and compliance. Communicate change management plans to address workforce concerns.	Platform and DevOps: Design modular agents and manage orchestration tools. Security: Implement API governance and monitor agent interactions. Integration teams: Adopt standards for interoperability and observability.	Process owners: Identify workflows for MAS pilots and validate outcomes. Finance: Manage unpredictable costs and fund observability tools. Operations: Support human-agent collaboration and training initiatives.

Внедрение DSLMs в 2026 - 2030 годах



Как и какой будет достигаться эффект

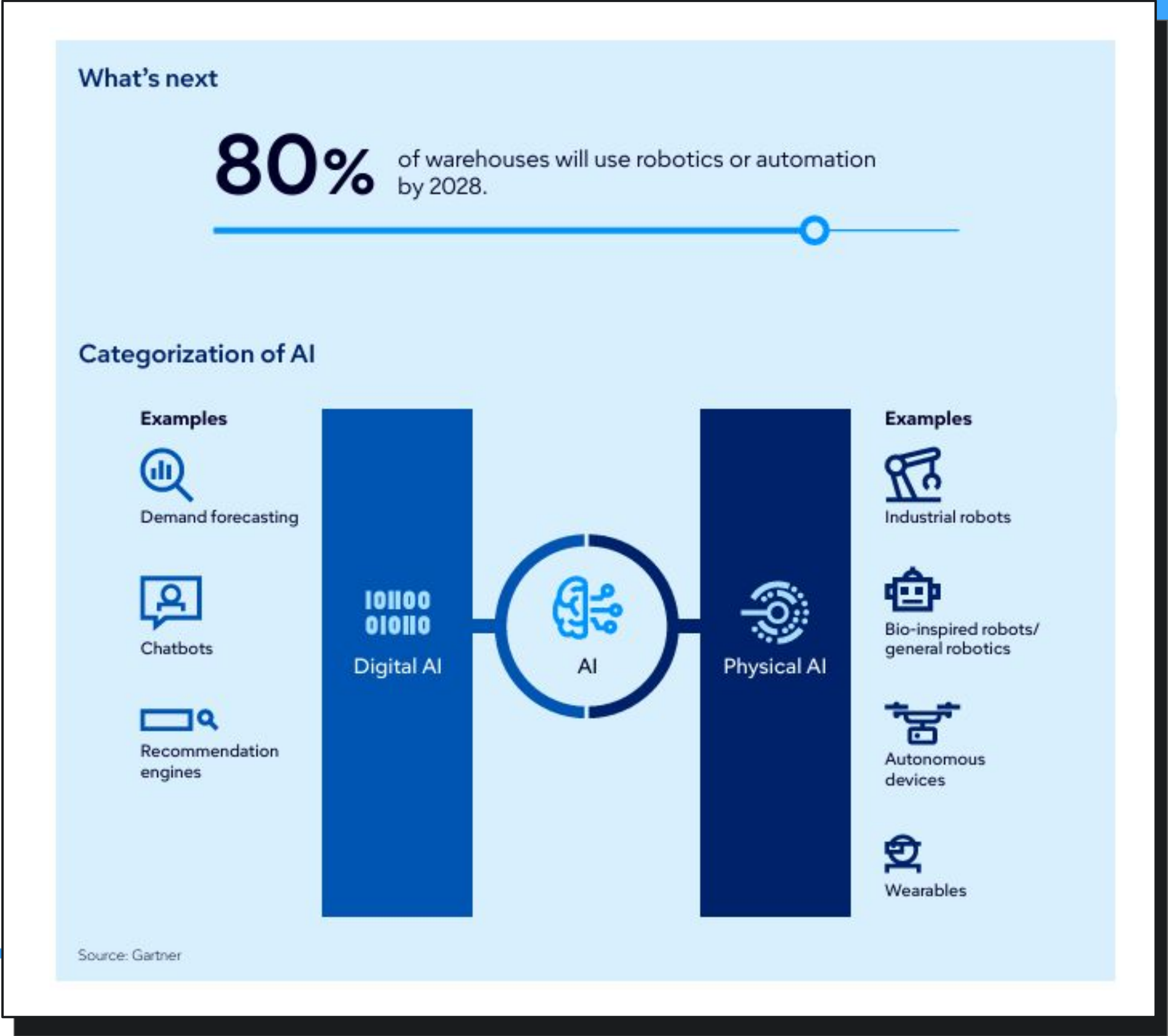
Action plan to deliver precise, industry-specific compliance

Steps	1 Identify high-impact use cases	2 Strengthen data governance	3 Pilot DSLMs in critical domains	4 Build cross-functional teams	5 Monitor and optimize
Expected outcome	Faster ROI and improved accuracy.	Reliable and compliant DSLM outputs.	Demonstrate measurable business value.	Smooth integration and adoption.	Sustainable performance and cost control.
Action	Target workflows where generic LLMs underperform.	Implement robust privacy and quality controls.	Start with finance, healthcare or HR processes.	Include IT, SMEs and compliance in DSLM projects.	Apply explainability and compliance frameworks.

Key players to support implementation success

CIO	IT partners	Business partners
Define DSLM strategy for regulated and high-value domains.	Data and analytics: Prepare domain-specific datasets and maintain quality.	Domain experts: Validate DSLM outputs for accuracy and relevance.
Ensure governance for accuracy, compliance and explainability.	ModelOps: Manage fine-tuning, monitoring and life cycle governance.	Finance: Budget for DSLM adoption and cost optimization.
Align DSLM adoption with ROI and risk management goals.	Security: Enforce privacy and compliance for DSLM deployments.	Compliance: Ensure adherence to regulatory standards.

Physical AI



Как и какой будет достигаться эффект

Action plan to automate real-world tasks and boost productivity everywhere

Steps	1 Audit operational domains	2 Pilot physical AI systems	3 Build cross-functional teams	4 Educate stakeholders	5 Plan for multiagent coordination
Expected outcome	Identify areas for automation and cost savings.	Validate performance and ROI.	Effective governance and integration.	Avoid confusion and misaligned investments.	Future-proof deployments.
Action	Target logistics, maintenance and safety workflows.	Use simulation and digital twins before live deployment.	Include IT, operations and engineering in planning.	Clarify distinctions between physical AI, embedded AI and edge AI.	Explore orchestration platforms for fleets of devices.

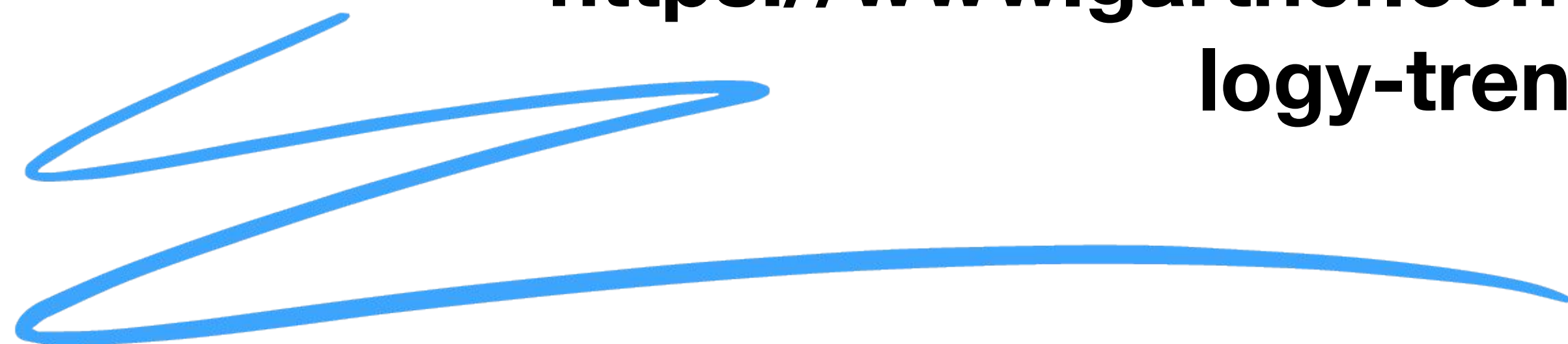
Key players to support implementation success

CIO	IT partners	Business partners
Define a physical AI strategy aligned with operational goals.	Infrastructure and operations: Integrate physical AI with IoT and legacy systems.	Operations: Identify high-value use cases and validate performance.
Ensure governance for safety, reliability and explainability.	Security: Implement safeguards for autonomous systems.	Finance: Budget for robotics and automation investments.
Partner with operations and engineering for integration and risk management.	Data and analytics: Support simulation and digital twin testing.	Compliance: Ensure adherence to safety and regulatory standards.

Где можно почитать всё исследование

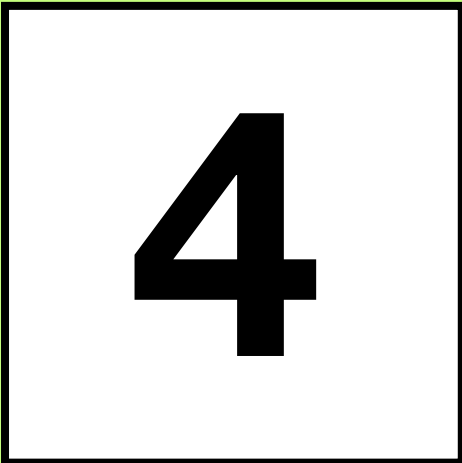
**Gartner_Top 10 Strategic Technology
Trends_2026 (29 pgs)**

<https://www.gartner.com/en/articles/top-technology-trends-2026>

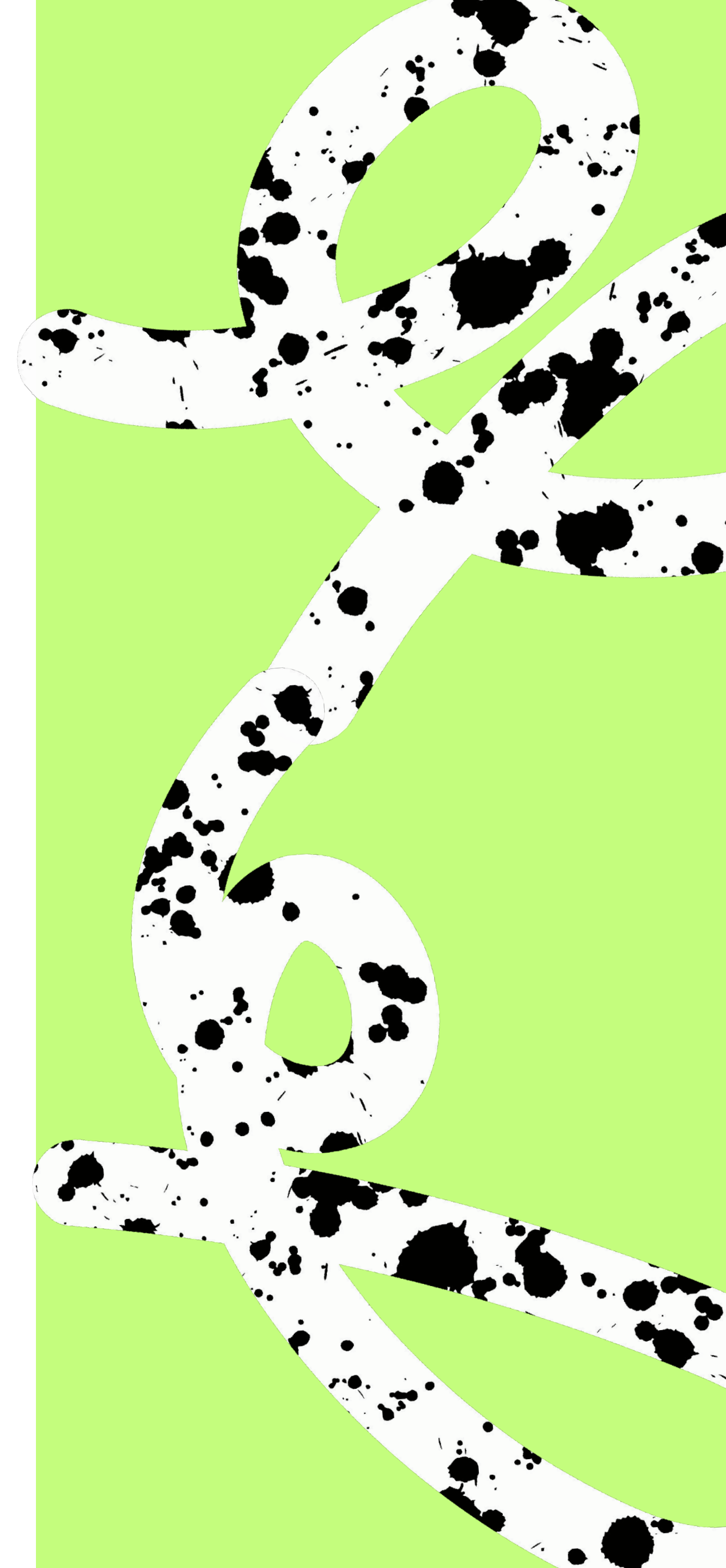


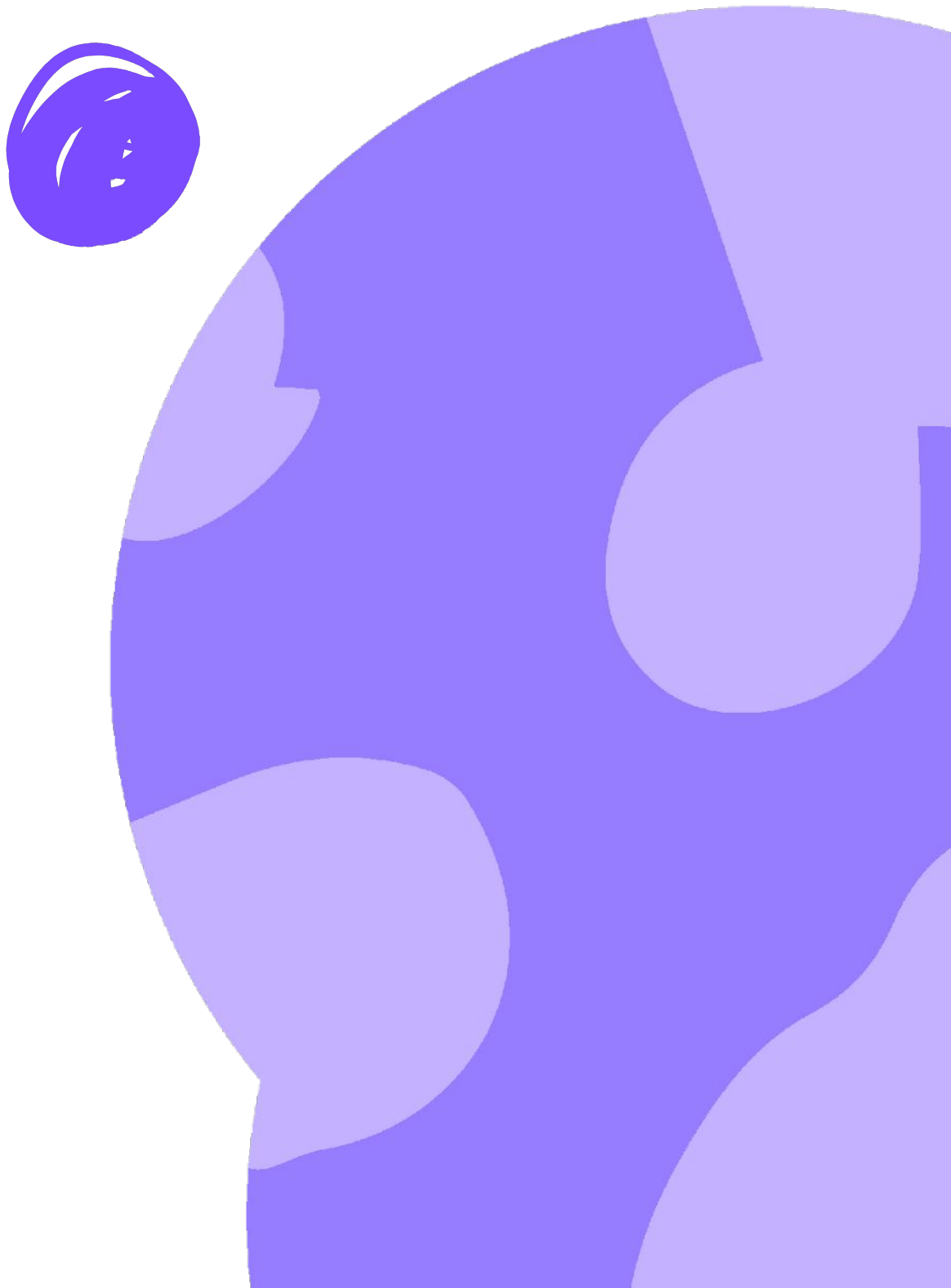
P3. Coding agents

1. Agent computer interface (ACI)

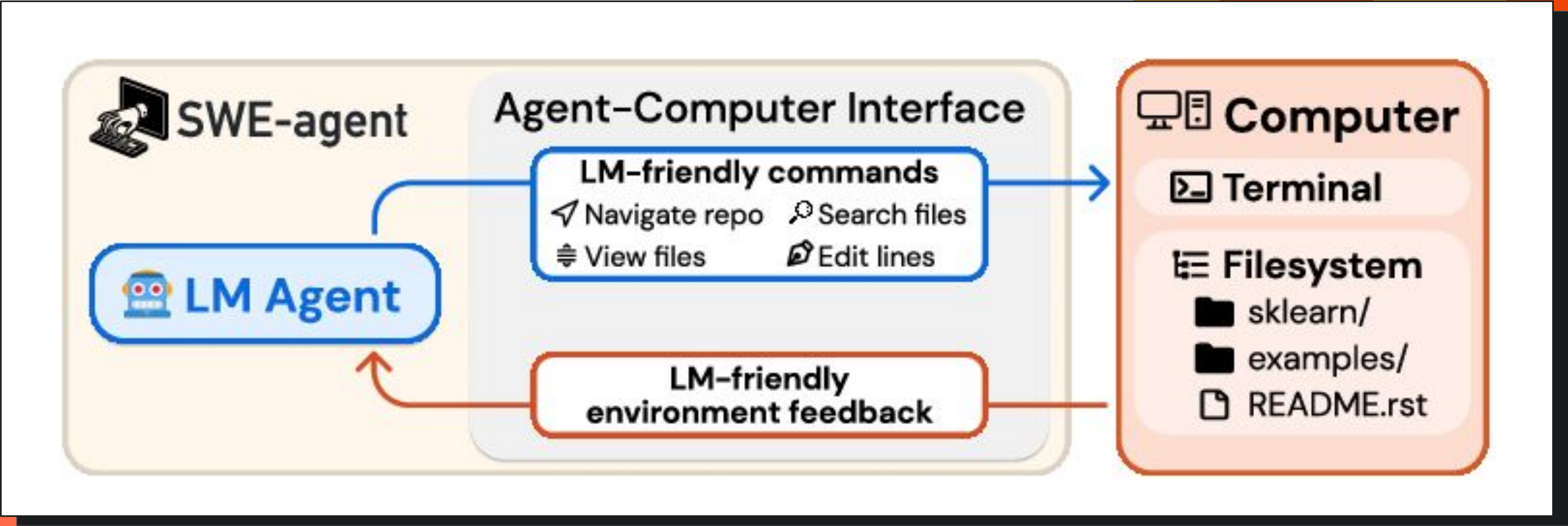


**Есть ли различие в архитектуре
с не LLM агентом?**



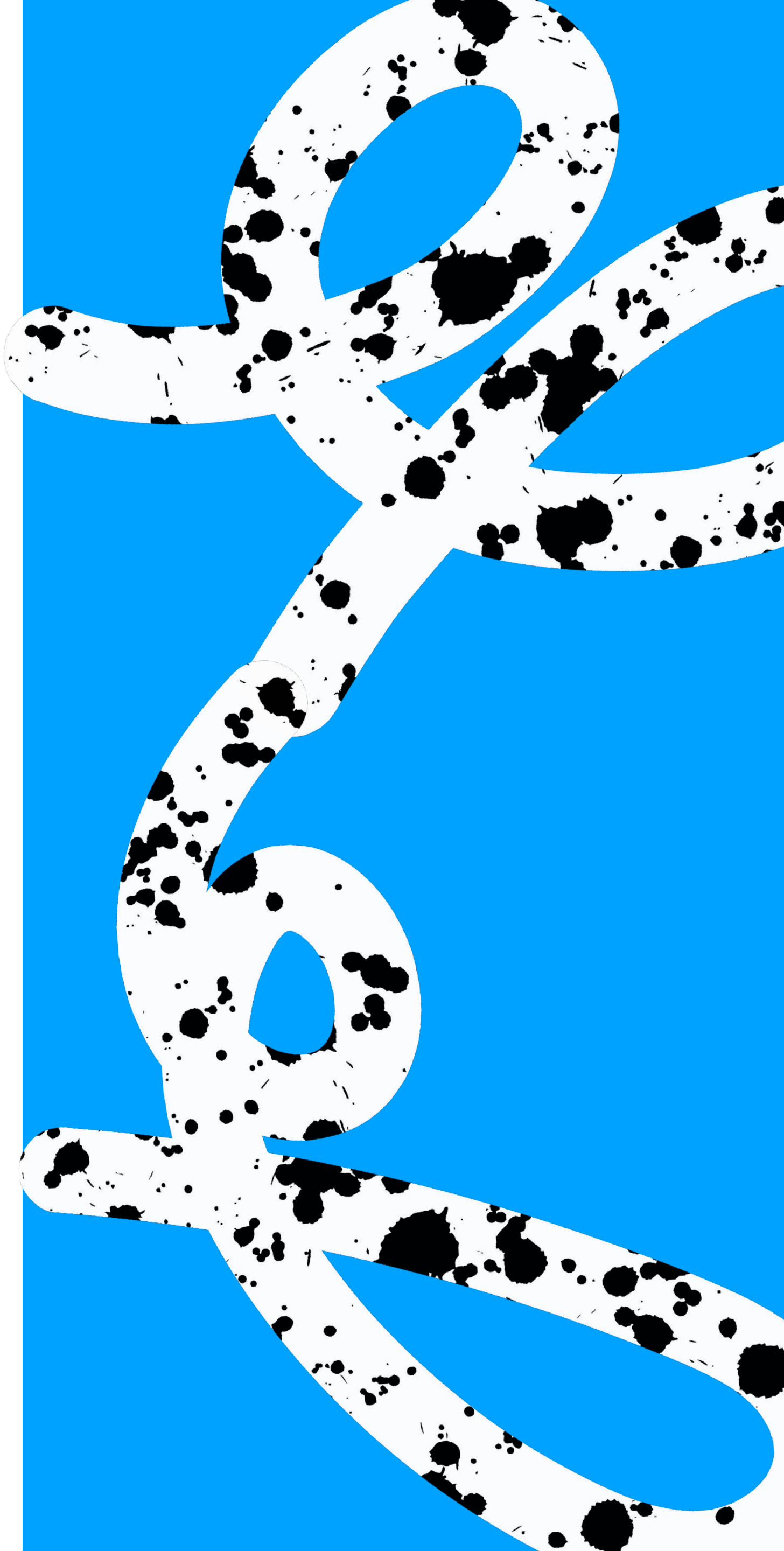


ACI



Arxiv

<https://arxiv.org/pdf/2405.15793>



Summary

1



2

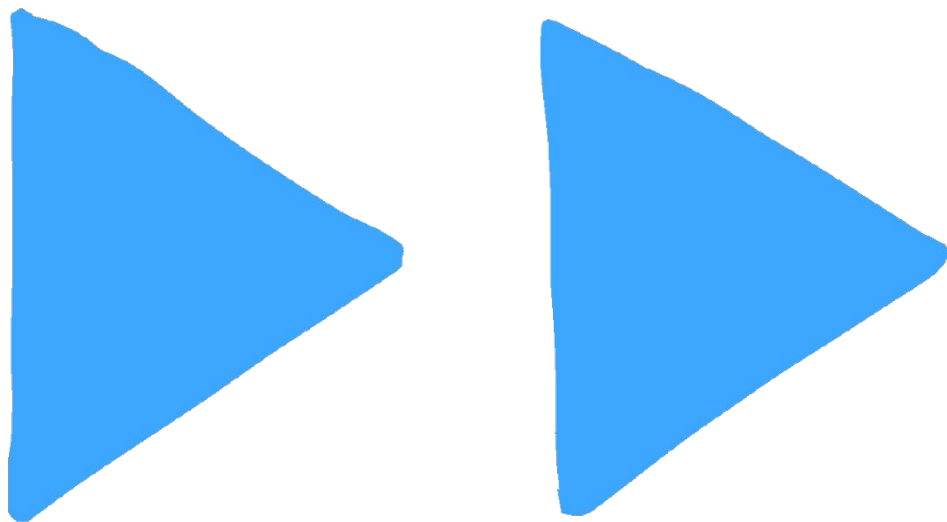
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4

Понятие “агент” существует уже 30+ лет.

Агента определяют:

- Память
- Планирование
- Реакция + Действие + инструменты.





**Поделитесь обратной
связью по занятию**

